Projective Gauge Reduction and Auxiliary Connection Elimination in a Metric-Affine Trace Sector

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Abstract

Independent affine connections with projective symmetry possess a gauge-degenerate auxiliary fibre, but gauge degeneracy need not exhaust all auxiliary degeneracy. We formulate a finite-dimensional local Hessian criterion that separates pure gauge null directions from additional invariant null directions. Applied to a four-dimensional metric-affine trace/vector model in which the compensator A_μ is varied, the criterion gives $\ker \mathcal{H}_{\text{full}} = \operatorname{im} G_x$ on the regular locus $f_1 \neq 0$, $f_2 \neq 0$. Consequently, the horizontal affine data have a unique quotient solution, the unreduced solutions form one local projective orbit, and quotient-first and eliminate-first reductions agree as spacetime-local bulk functionals modulo the admitted divergence currents. On the singular strata $f_1 = 0$ or $f_2 = 0$, rank loss can instead produce source incompatibility or gauge-invariant algebraic moduli. The result is local to the stated trace/vector configuration space and identifies the geometric condition under which projectively degenerate auxiliary connection elimination is quotient independent. Within a two-parameter symmetry-compatible deformation family, the same analysis yields an explicit shifted Schur regularity surface and a source-compatibility condition on its rank-changing locus.

Keywords: metric-affine geometry; projective symmetry; Palatini gravity; gauge reduction; auxiliary fields; Hessian degeneracy

1 Introduction

When does elimination of a gauge-degenerate auxiliary field descend to a well-defined operation on the gauge quotient?

An independent affine connection in Palatini and metric-affine geometry is typically auxiliary, and the Palatini curvature scalar is invariant under a consistent projective shift [1–6]. Neither projective invariance nor algebraic connection elimination is claimed as new [7–12]. The geometric issue is whether elimination is unique on the quotient and whether quotienting commutes with elimination.

Gauge nonuniqueness and auxiliary nonuniqueness need not coincide. Raw affine nonuniqueness splits into two distinct effects. Some directions are vertical tangent directions to the projective orbit and are removed by the quotient. Other null directions can survive in the invariant horizontal data. Gauge invariance is not dynamical independence, and gauge redundancy is not the same as auxiliary eliminability. The auxiliary Hessian therefore supplies the local geometric test: its kernel must contain the gauge image, but a unique quotient solve requires that it contain nothing more.

Independent affine connections are commonly eliminated algebraically in Palatini and metric-affine analyses [4, 5, 7–11, 13]. If the affine Hessian has gauge-invariant null directions beyond the projective orbit, that substitution does not define a unique quotient theory:

unresolved invariant data remain. The exact kernel criterion is therefore a local diagnostic of when integrating out the connection is a quotient-independent operation rather than an implicit non-gauge branch restriction.

Ferrara, Guzmán, and Järv study generalized projective symmetry, connection-operator singularity, symmetry-generated and accidental zero modes, source constraints, symmetry-adapted solution methods, and symmetry breaking in broader quadratic metric-affine gravity [14]. None of those general structures is claimed as new here. For the specified compensator-extended trace/vector auxiliary system, the present paper instead gives a closed evaluation of the full one-form-fibre kernel, quotient eliminability and local orbit, an explicit quotient/elimination bulk comparison, undeformed singular-source compatibility, and the shifted Schur/source conditions in a controlled invariant deformation family.

The trace/vector sector is a minimal nontrivial setting in this model where a projective one-form orbit, gauge-invariant horizontal affine data, and rank-changing algebraic connection equations coexist. It is therefore an exact laboratory for separating projective gauge degeneracy from invariant failure of auxiliary eliminability. This defined configuration space is not asserted to be a dynamically consistent truncation of unrestricted 64-component metric-affine gravity.

On the regular locus of the model below, the vertical projective direction is quotiented and the horizontal affine variables are uniquely eliminated. Beyond it, extra gauge-invariant kernel can leave unresolved invariant data rather than additional projective freedom. The algebraic source conditions are sharp: S2 ($f_1 = 0$, $f_2 \neq 0$) requires $\partial f_1 = 0$; S3 ($f_1 \neq 0$, $f_2 = 0$) requires $\partial f_3 = 0$; and S4 ($f_1 = f_2 = 0$) requires both. Incompatible sources give no stationary affine solution, while compatible branches can retain invariant moduli. These statements classify algebraic branches, not dynamical crossings.

Earlier public companion work studies the same compensated trace-residue chart with A_μ held as a nondynamical reference on the projective fibre and establishes the local q/u Schur decoupling and strict residue-removal classification in that posture [15]. The present paper changes the variational problem by including A_μ in the auxiliary configuration space. Its Euler–Lagrange equation is projectively Noether-redundant rather than an independent physical equation; varying A_μ makes the full projective auxiliary fibre explicit. The full affine species Hessian, the true one-form-fibre kernel, quotient uniqueness, singular-source compatibility, and the quotient-first versus eliminate-first comparison are consequently additional variational questions not contained in the nondynamical-A_μ classification.

Section 2 records a finite-dimensional organizing criterion for quadratic auxiliary densities. It is a reduction lemma, not a deep new general mathematical theorem. Sections 3–6 supply the principal scientific content: a nontrivial verification of its hypotheses in a four-dimensional metric-affine trace/vector model. The Palatini kinetic density $3(\partial f_1)^2/(2f_1)$ is generic structure [4, 7, 8]. In this varied-A_μ system, the model-specific contribution is the full affine Hessian and its exact full-fibre gauge kernel, the sharp regular-locus local orbit, explicit equality of quotient-first and eliminate-first reductions, and singular-stratum source compatibility. The transverse determinant $-f_1^2 f_2/6$ supplies the regularity test for those results. The kernel/source-compatibility mechanism is also tracked across a two-parameter projectively invariant quadratic deformation family, where the regularity boundary becomes an explicit Schur surface.

Main results and geometric significance. On the undeformed regular locus, $\ker \mathcal{H}_{\text{full}} = \operatorname{im} G_x$, while $\det H_\perp = -f_1^2 f_2/6$ separates the unavoidable vertical gauge degeneracy from horizontal auxiliary regularity. In the controlled deformation family the horizontal rank-changing boundary moves to $S_t(\alpha, \beta) = 0$. Thus projective symmetry fixes the vertical kernel, whereas invariant coupling structure decides whether an additional horizontal kernel appears; at rank loss the outcome can be source incompatibility or invariant moduli rather than extra gauge freedom.

2 A finite-dimensional reduction criterion

Let E be a finite-dimensional auxiliary fibre over spectator fields ψ, with coordinates X^A and bulk density

$$L_{\mathrm{aux}}(X;\psi) = \tfrac{1}{2}X^A H_{AB}(\psi)X^B + J_A(\psi)X^A + L_0(\psi), \tag{2.1}$$

H symmetric. Let $G : \mathfrak{g} \to E$ be linear and injective, acting only in the fibre: $\delta_\varepsilon X = G\varepsilon$.

Gauge invariance $L_{\mathrm{aux}}(X + G\varepsilon) = L_{\mathrm{aux}}(X)$ for all X, ε expands to

$$HG = 0, \qquad G^T J = 0. \tag{2.2}$$

No further algebraic condition follows from invariance of (2.1).

Theorem 2.1 (Finite-dimensional gauge-auxiliary reduction criterion)**.** *Assume* (2.2) *and* $\ker H = \operatorname{im} G$*. Equivalently, in finite dimensions, the induced map* $\overline{H} : E/\operatorname{im} G \to (E/\operatorname{im} G)^*$ *is invertible. Then:*

1. $HX + J = 0$ *has a unique solution class in* $E/\operatorname{im} G$*;*
2. *the unreduced solution set is one affine gauge orbit* $X_\star + \operatorname{im} G$*;*
3. *the on-shell value* $L_{\mathrm{on}} = L_0 + \frac{1}{2}J^T X$ *is constant on that orbit;*
4. *quotient-first elimination and eliminate-first quotient produce the same local on-shell functional;*
5. *any two linear horizontal charts related by* $\mathrm{GL}(n)$ *have the same regular locus and the same on-shell value.*

Proof. Invariance gives $\operatorname{im} G \subseteq \ker H$; the kernel hypothesis gives equality. Solvability of $HX = -J$ is $J \in (\operatorname{im} G)^0$, which is $G^T J = 0$. The solution space is one affine orbit of direction $\ker H$. Finite-dimensional nondegeneracy of $\overline{H}$ yields a unique class $[X_\star]$. On shell, $L_{\mathrm{on}} = L_0 + \frac{1}{2}J^T X$, and $J^T G = 0$ makes this representative-independent. Both reduction orders substitute the same class into (2.1). A $\mathrm{GL}(n)$ change of horizontal coordinates multiplies the determinant of the induced transverse form $H_\perp$ by a nowhere-zero squared Jacobian, leaving the regular locus and the on-shell scalar L_{on} invariant. □

Theorem 2.1 provides an organizing finite-dimensional criterion for a quadratic form with a prescribed kernel. Sufficiency is all that is used below. It does not replace a model Hessian computation.

In a splitting $E = \mathcal{H} \oplus \operatorname{im} G$ with coordinates (x, V),

$$H = \begin{pmatrix} H_\perp & 0 \\ 0 & 0 \end{pmatrix}, \tag{2.3}$$

and $\ker H = \operatorname{span}\{e_V\}$ precisely when $\det H_\perp \neq 0$, with $x_\star = -H_\perp^{-1} J_{\mathcal{H}}$ and $L_{\mathrm{on}} = L_0 - \frac{1}{2}J_{\mathcal{H}}^T H_\perp^{-1} J_{\mathcal{H}}$.

Remark 2.2 (Species matrices and the one-form fibre). If the auxiliary fibre at a spacetime point x is of the form $E_x = S \otimes T_x^* M$ and the quadratic density is built from a species matrix H_{AB} contracted with a nondegenerate spectator metric, the true Hessian is $H \otimes g^{-1}$. Then

$$\ker(H \otimes g^{-1}) = \ker H \otimes T_x^* M, \qquad \operatorname{rank}(H \otimes g^{-1}) = D \operatorname{rank}_{\mathrm{sp}}(H). \tag{2.4}$$

Species invertibility and kernel criteria therefore lift directly to the full one-form fibre. Theorem 2.1 is applied below to that full pointwise fibre; the displayed 3×3 and 4×4 matrices are the species coefficients.

Quotient geometry. At each spacetime point the full auxiliary fibre fits into the short exact sequence

$$0 \longrightarrow \operatorname{im} G_x \longrightarrow E_{\mathrm{full},x} \longrightarrow E_{\mathrm{full},x}/\operatorname{im} G_x \longrightarrow 0.$$

Projective invariance gives $\operatorname{im} G_x \subseteq \ker \mathcal{H}_{\mathrm{full}}$, so the Hessian descends to a bilinear form on the quotient. The equality $\ker \mathcal{H}_{\mathrm{full}} = \operatorname{im} G_x$ is equivalent to nondegeneracy of that quotient form; when equality fails, $\ker \mathcal{H}_{\mathrm{full}}/\operatorname{im} G_x$ is precisely the additional gauge-invariant degeneracy. This is a geometric interpretation of the criterion, not a new general theorem.

Uniqueness of the reduced solve for *every* compatible source requires $\ker H = \operatorname{im} G$. If $\ker H$ strictly contains $\operatorname{im} G$, the quotient retains additional gauge-invariant degeneracies; selecting a representative of those directions requires a non-gauge restriction, branch choice, or additional dynamics, and is not a gauge quotient. Equality of on-shell *values* for a single compatible source can still occur; unrestricted necessity of QE = EQ is not claimed. The criterion is fibrewise linear algebra. It is not a symplectic reduction theorem [12, 16, 17].

3 Metric-affine trace model

Spacetime dimension $D = 4$, signature $(-,+,+,+)$. The curvature, torsion, and nonmetricity conventions are

$$R^{\lambda}{}_{\rho\mu\nu} = \partial_\mu \Gamma^{\lambda}{}_{\nu\rho} - \partial_\nu \Gamma^{\lambda}{}_{\mu\rho} + \Gamma^{\lambda}{}_{\mu\sigma}\Gamma^{\sigma}{}_{\nu\rho} - \Gamma^{\lambda}{}_{\nu\sigma}\Gamma^{\sigma}{}_{\mu\rho}, \tag{3.1}$$

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}, \qquad R = g^{\mu\nu}R_{\mu\nu}, \tag{3.2}$$

$$T^{\lambda}{}_{\mu\nu} = 2\Gamma^{\lambda}{}_{[\mu\nu]}, \qquad T_\mu = T^{\lambda}{}_{\lambda\mu}, \tag{3.3}$$

$$Q_{\lambda\mu\nu} = -\nabla_\lambda g_{\mu\nu}, \qquad Q_\lambda = g^{\mu\nu}Q_{\lambda\mu\nu}, \qquad \widetilde{Q}_\nu = g^{\mu\lambda}Q_{\lambda\mu\nu}. \tag{3.4}$$

The affine derivative on the metric is

$$\nabla_\lambda g_{\mu\nu} = \partial_\lambda g_{\mu\nu} - \Gamma^{\rho}{}_{\lambda\mu}g_{\rho\nu} - \Gamma^{\rho}{}_{\lambda\nu}g_{\mu\rho}. \tag{3.5}$$

The affine configuration space of the model is the three traces $(T_\mu, Q_\mu, \widetilde{Q}_\mu)$ together with a compensator one-form A_μ. Axial torsion, tensor torsion, and traceless nonmetricity exist as metric-affine irreps [1, 18] and are not coordinates here: they are neither solved to zero nor removed by the projective gauge. Matter is metric-minimal, so no hypermomentum sources the connection [19, 20].

Assumption 3.1 (Trace/vector configuration space)**.** The independent affine data are $(T, Q, \widetilde{Q}, A)$. Deleted irreps remain outside the domain.

Assumption 3.2 (Metric-minimal matter)**.** Matter does not supply an independent hypermomentum source for the connection equation.

$$\bar{T}_\mu := T_\mu - A_\mu. \tag{3.6}$$

The residue $t_\mu := \bar{T}_\mu$ is the projectively invariant torsion-trace combination. The bulk action is

$$S = \int \mathrm{d}^4x\, \sqrt{-g}\, \Big[f_1\, R(\Gamma) + f_2\, \bar{T}^2 + f_3\, \bar{\nabla}\cdot\bar{T} + U(\phi) \Big], \tag{3.7}$$

with coefficient functions of a scalar argument ϕ. This is a representative, compact, nontrivial projectively invariant trace-residue model: $f_1 R(\Gamma)$ supplies the projective gauge and the Palatini block, while $f_2\bar{T}^2 + f_3\bar{\nabla}\cdot\bar{T}$ keep a gauge-invariant affine residue algebraically active. It is not claimed to be the most general metric-affine Lagrangian, nor a unique minimal operator basis.

Assumption 3.3 (Trace-residue action)**.** The bulk functional is (3.7). The potential $U(\phi)$ is affine-independent.

The one-form A_μ is introduced as a compensator field for the one-form projective action. No microscopic or ultraviolet Stueckelberg interpretation is assumed. The combination T^2 is not projectively invariant, while $\bar{T}^2$ is, once $\delta A = 3\xi$ accompanies the connection shift. In the gauge-adapted coordinates of Section 4, $V_\mu = A_\mu$ parametrizes the projective orbit, while $t_\mu = T_\mu - A_\mu$ is gauge invariant but remains auxiliary on $U_{\rm reg}$. Promoting A_μ from a nondynamical reference to a varied auxiliary variable is not claimed to introduce a propagating degree of freedom; it changes the auxiliary stationarity problem. The resulting question is whether the full Hessian kernel contains only the projective vertical direction or also invariant null directions. After the quotient, A_μ parametrizes the orbit rather than an independent bulk degree of freedom.

4 Projective transformation and invariant coordinates

The projective transformation used throughout is

$$\delta\Gamma^\lambda{}_{\mu\nu} = -\delta^\lambda{}_\nu\xi_\mu, \qquad \delta A_\mu = 3\xi_\mu, \tag{4.1}$$

with $\delta g_{\mu\nu} = 0$. Index placement is part of the law under the targets below. The MAG first-kind shift $\delta\Gamma^\lambda{}_{\mu\nu} = +\delta^\lambda{}_\mu\xi_\nu$ [1, 6] is a different transformation, not used here.

Linear vectorial connection maps form a larger family [6, 21]. The two-parameter subfamily containing common first- and last-index writings [2, 3, 22] is

$$\delta\Gamma^\lambda{}_{\mu\nu} = a\,\delta^\lambda{}_\mu\xi_\nu + b\,\delta^\lambda{}_\nu\xi_\mu, \tag{4.2}$$

with compensator increment held at

$$\delta A_\mu = +3\xi_\mu. \tag{4.3}$$

At $D = 4$,

$$\delta T_\mu = 3(a-b)\xi_\mu, \qquad \delta Q_\mu = (2a+8b)\xi_\mu, \qquad \delta\widetilde{Q}_\mu = (5a+2b)\xi_\mu. \tag{4.4}$$

The linear Ricci shift is

$$\delta R_{\alpha\beta}\big|_\partial = b\,\partial_\alpha\xi_\beta + (-3a-b)\,\partial_\beta\xi_\alpha. \tag{4.5}$$

The quadratic remainder in the connection difference vanishes if and only if $a = 0$. The Ricci target

$$\delta R_{\mu\nu} = -2\partial_{[\mu}\xi_{\nu]}. \tag{4.6}$$

together with $\delta T = +3\xi$ forces

$$a = 0, \qquad b = -1. \tag{4.7}$$

Proposition 4.1 (Selection within the adopted two-parameter connection-shift family)**.** *Under the conventions of Section 3 and the simultaneous targets*

$$\delta T = +3\xi, \qquad \delta A = +3\xi, \qquad \delta(T-A) = 0, \qquad \delta R = 0, \qquad \delta R_{\mu\nu} = -2\partial_{[\mu}\xi_{\nu]}, \tag{4.8}$$

the family (4.2) *admits the unique solution* $a = 0$, $b = -1$. *Equivalently, the law is* (4.1). *A constant rescaling of* ξ *reparametrizes the same law.*

The induced shifts are

$$\delta(T_\mu, Q_\mu, \widetilde{Q}_\mu, A_\mu) = (3, -8, -2, 3)\xi_\mu. \tag{4.9}$$

Hence $\delta\bar{T} = 0$ and

$$\delta R^\lambda{}_{\rho\mu\nu} = -\delta^\lambda{}_\rho\, 2\partial_{[\mu}\xi_{\nu]}, \tag{4.10}$$

so $\delta R = 0$. The first-kind pair $(a, b) = (1, 0)$ yields

$$\delta(T, Q, \widetilde{Q}) = (3, 2, 5)\xi \tag{4.11}$$

and fails (4.6); it is excluded only under these targets.

Write

$$\Gamma^\lambda{}_{\mu\nu} = \Gamma^\lambda_{\mathrm{LC}\,\mu\nu} + N^\lambda{}_{\mu\nu}. \tag{4.12}$$

The trace-sector distortion is

$$\begin{aligned} N^\lambda{}_{\mu\nu} = {} & \tfrac{1}{3}\big(\delta^\lambda{}_\mu T_\nu - g_{\mu\nu}T^\lambda\big) \\ & + \tfrac{1}{36}\big(5\delta^\lambda{}_\mu Q_\nu + 5\delta^\lambda{}_\nu Q_\mu - 7g_{\mu\nu}Q^\lambda\big) \\ & + \tfrac{1}{18}\big(-\delta^\lambda{}_\mu \widetilde{Q}_\nu - \delta^\lambda{}_\nu \widetilde{Q}_\mu + 5g_{\mu\nu}\widetilde{Q}^\lambda\big), \end{aligned} \tag{4.13}$$

Invariant coordinates adapted to (4.1) are

$$t_\mu = T_\mu - A_\mu, \qquad q_\mu = Q_\mu + \tfrac{8}{3}A_\mu, \qquad u_\mu = \widetilde{Q}_\mu + \tfrac{2}{3}A_\mu, \qquad V_\mu = A_\mu, \tag{4.14}$$

with inverse

$$T = t + V, \qquad Q = q - \tfrac{8}{3}V, \qquad \widetilde{Q} = u - \tfrac{2}{3}V, \qquad A = V. \tag{4.15}$$

Thus $t = \bar{T}$.

Lemma 4.2 (Horizontal invariance)**.** *Under* (4.9)*, $\delta t = \delta q = \delta u = 0$ and $\delta V = 3\xi$. The coordinate $V_\mu = A_\mu$ is vertical; (q, u, t) are invariant.*

Assumption 4.3 (Regular locus)**.** Invertibility, orbit, and reduced-action statements are made on

$$U_{\mathrm{reg}} := \{f_1 \neq 0\} \cap \{f_2 \neq 0\}. \tag{4.16}$$

Assumption 4.4 (Local bulk identification)**.** The theorem is spacetime-local on open regular regions. Bulk densities are identified modulo the admitted total divergences generated by the currents J_1 and J_3 of Appendix H. That identification is a variational-boundary convention. It does not classify proper versus improper projective transformations.

Signs $f_1 > 0$, $f_2 > 0$, and $K_{\mathrm{eff}} > 0$ are not hypotheses.

5 Exact transverse Hessian

5.1 Exact trace expansion

Insert (4.13) into the connection-difference identity for (3.1) and contract to the scalar curvature. Every quadratic and divergence coefficient is determined. There is no remainder in the trace sector:

$$\begin{aligned} R - R_{\mathrm{LC}} = {} & \bar{\nabla}_\mu\big(-2T^\mu - Q^\mu + \widetilde{Q}^\mu\big) \\ & - \tfrac{2}{3}T^2 - \tfrac{11}{72}Q^2 + \tfrac{1}{18}\widetilde{Q}^2 - \tfrac{2}{3}T\cdot Q + \tfrac{2}{3}T\cdot\widetilde{Q} + \tfrac{2}{9}Q\cdot\widetilde{Q}. \end{aligned} \tag{5.1}$$

Here $X \cdot Y := g^{\mu\nu}X_\mu Y_\nu$ and $X^2 := X \cdot X$. The six quadratic coefficients are

$$\Big(-\tfrac{2}{3},\ -\tfrac{11}{72},\ +\tfrac{1}{18},\ -\tfrac{2}{3},\ +\tfrac{2}{3},\ +\tfrac{2}{9}\Big) \tag{5.2}$$

in the order $(T^2, Q^2, \widetilde{Q}^2, T\cdot Q, T\cdot\widetilde{Q}, Q\cdot\widetilde{Q})$. Equation (5.1) is the complete expansion used below; no ellipsis is implied. The derivation is given in Appendix C.

Table 1: Conventions and projective transformations at $D = 4$.

Object	Definition or law	Variation
$R^{\lambda}{}_{\rho\mu\nu}$	(3.1)	$-\delta^{\lambda}{}_{\rho}\, 2\partial_{[\mu}\xi_{\nu]}$
$R_{\mu\nu}$	(3.2)	$-2\partial_{[\mu}\xi_{\nu]}$
R	$g^{\mu\nu}R_{\mu\nu}$	0
T_μ	(3.3)	$+3\xi_\mu$
Q_μ	(3.4)	$-8\xi_\mu$
$\widetilde{Q}_\mu$	(3.4)	$-2\xi_\mu$
A_μ	compensator	$+3\xi_\mu$
$\bar{T}_\mu$	$T_\mu - A_\mu$	0
$\Gamma^{\lambda}{}_{\mu\nu}$	independent connection	$-\delta^{\lambda}{}_{\nu}\xi_\mu$

5.2 Projective cancellation in raw traces

Substituting the projective shifts (4.9) into the divergence and into the quadratic polynomial of (5.1) cancels all vertical dependence. Equivalently, after (4.15) both the divergence current

$$W^\mu := -2T^\mu - Q^\mu + \widetilde{Q}^\mu \tag{5.3}$$

and the quadratic form become functions of (t, q, u) only:

$$W^\mu = -2t^\mu - q^\mu + u^\mu, \tag{5.4}$$

and the quadratic polynomial is that of (5.1) with $(T, Q, \widetilde{Q})$ replaced by (t, q, u). A constant pure vertical connection has vanishing scalar curvature, which is an independent control of $\delta R = 0$.

5.3 Integration by parts

Multiply (5.1) by f_1 and use the two bulk identities

$$\sqrt{-g}\, f_1\, \bar{\nabla}_\mu W^\mu = -\sqrt{-g}\, (\partial_\mu f_1) W^\mu + \partial_\mu\big(\sqrt{-g}\, f_1 W^\mu\big), \tag{5.5}$$

$$\sqrt{-g}\, f_3\, \bar{\nabla}_\mu \bar{T}^\mu = -\sqrt{-g}\, (\partial_\mu f_3) \bar{T}^\mu + \partial_\mu\big(\sqrt{-g}\, f_3 \bar{T}^\mu\big). \tag{5.6}$$

The currents actually generated are

$$J_1^\mu = f_1 W^\mu, \qquad J_3^\mu = f_3 \bar{T}^\mu. \tag{5.7}$$

Bulk functionals that differ by $\partial_\mu\big(\sqrt{-g}\, (c_1 J_1^\mu + c_3 J_3^\mu)\big)$ are identified. That is the admitted variational-boundary class; it is not a classification of all possible surface terms, and it does not distinguish proper from improper projective transformations (Appendix H).

5.4 Horizontal affine quadratic form

After (5.5)–(5.6) and the horizontal rewrite, the affine bulk density (omitting $f_1 R_{\mathrm{LC}} + U(\phi)$ and omitting the admitted divergence) is quadratic-plus-linear in the species triple $x := (q, u, t)^{\mathsf{T}}$:

$$\mathcal{L}_{\mathrm{aff}} = \tfrac{1}{2} x^{\mathsf{T}} H_\perp x + J^{\mathsf{T}} x. \tag{5.8}$$

Each species coordinate is a one-form. The displayed 3×3 matrix $H_\perp$ is a *species-space* coefficient matrix: the quadratic pairing is the metric contraction $X^A \cdot Y^B := g^{\mu\nu} X^A_\mu Y^B_\nu$. At each spacetime point x write $S_\perp \simeq \mathbb{R}^3$ and

$$E_{\perp,x} = S_\perp \otimes T^*_x M. \tag{5.9}$$

The true pointwise transverse Hessian is therefore

$$\mathcal{H}_\perp = H_\perp \otimes g^{-1}. \tag{5.10}$$

Because g^{-1} is nondegenerate,

$$\ker \mathcal{H}_\perp = \ker H_\perp \otimes T_x^* M, \qquad \operatorname{rank} \mathcal{H}_\perp = D \operatorname{rank}_{\mathrm{sp}}(H_\perp). \tag{5.11}$$

The species matrix and source are

$$H_\perp = f_1 H_0 + \operatorname{diag}(0, 0, 2f_2), \tag{5.12}$$

$$H_0 = \begin{pmatrix} -11/36 & 2/9 & -2/3 \\ 2/9 & 1/9 & 2/3 \\ -2/3 & 2/3 & -4/3 \end{pmatrix}, \tag{5.13}$$

$$J = \big(\partial f_1,\ -\partial f_1,\ 2\partial f_1 - \partial f_3\big)^{\mathsf{T}}. \tag{5.14}$$

There is no V dependence. The matrix H_0 is the Hessian of the quadratic polynomial in (5.1) with respect to (q, u, t). The diagonal $2f_2$ is the Hessian contribution of $f_2 t^2$. The source (5.14) is the linear form generated by (5.5)–(5.6). Detailed contractions are in Appendices C and D.

The two nonmetricity traces (q, u) enter $H_\perp$ through the upper-left 2×2 block. Isolating that block is the Palatini step: it eliminates the connection data that mix with f_1 before the residue equation is solved. The existence of such a Palatini block is generic [4, 5, 8]; the numerical matrix below is convention-specific.

5.5 The exact matrix

Write $q_N := (q, u)^{\mathsf{T}}$. Then

$$P := f_1 \begin{pmatrix} -11/36 & 2/9 \\ 2/9 & 1/9 \end{pmatrix}. \tag{5.15}$$

A direct determinant computation gives

$$\det P = f_1^2 \Big(\big(-\tfrac{11}{36}\big)\big(\tfrac{1}{9}\big) - \big(\tfrac{2}{9}\big)^2 \Big) = f_1^2 \Big(-\tfrac{11}{324} - \tfrac{4}{81} \Big) = f_1^2 \Big(-\tfrac{11}{324} - \tfrac{16}{324} \Big) = -\frac{f_1^2}{12}. \tag{5.16}$$

Hence P is invertible if and only if $f_1 \neq 0$. The inverse is

$$P^{-1} = f_1^{-1} \begin{pmatrix} -4/3 & 8/3 \\ 8/3 & 11/3 \end{pmatrix}. \tag{5.17}$$

No further field-dependent factor occurs in $\det P$. In particular the schematic notations M_{Pal} and Palatini_response are replaced, for all subsequent statements, by (5.15)–(5.17).

Lemma 5.1 (Palatini invertibility)**.** *On the present domain,* $\operatorname{rank}_{\mathrm{sp}}(P) = 2$ *if and only if* $f_1 \neq 0$*. At* $f_1 = 0$ *one has* $P = 0$*, so the Palatini pair cannot be solved.*

5.6 Explicit q/u solve

At fixed residue t, the (q, u) Euler–Lagrange equation is

$$P\, q_N + f_1 \begin{pmatrix} -2/3 \\ 2/3 \end{pmatrix} t + \begin{pmatrix} \partial f_1 \\ -\partial f_1 \end{pmatrix} = 0. \tag{5.18}$$

Applying (5.17) yields

$$q = 4\,\frac{\partial f_1}{f_1} - \frac{8}{3} t, \qquad u = \frac{\partial f_1}{f_1} - \frac{2}{3} t, \tag{5.19}$$

on $f_1 \neq 0$. The algebra is recorded in Appendix D. Equation (5.19) is the complete Palatini response in the horizontal chart: there is no unspecified matrix standing between the source and (q, u).

The same inverse applied to the f_1-sourced part alone produces the Palatini particular solution

$$(q, u)\big|_{t=0} = \big(4r,\, r\big), \qquad r := \frac{\partial f_1}{f_1}, \tag{5.20}$$

which will combine with the residue solution in the next section.

With (q, u) eliminated by (5.19), the remaining horizontal unknown is the invariant residue $t = \bar{T}$. The reduced t-problem is the Schur complement of P in $H_\perp$.

5.7 Exact Schur data

A direct complement computation (Appendix E) produces

$$S_t = 2f_2, \qquad J_{t,\mathrm{eff}} = -\partial f_3, \qquad \mathcal{L}_{\mathrm{const}} = \frac{3(\partial f_1)^2}{2f_1}. \tag{5.21}$$

The on-shell Palatini contribution is therefore the constant (in t) density $3(\partial f_1)^2/(2f_1)$, and the remaining quadratic-plus-linear t-density is

$$f_2\, t^2 - (\partial f_3) \cdot t. \tag{5.22}$$

5.8 Vanishing of the induced residue couplings

Write the would-be induced couplings after Palatini elimination as

$$\widehat{f_2} = f_2 + \Delta_2[f_1], \qquad \widehat{f_3} = f_3 + \Delta_3[f_1]. \tag{5.23}$$

Comparing (5.21)–(5.22) with the original residue terms in (3.7) gives

$$\Delta_2[f_1] = 0, \qquad \Delta_3[f_1] = 0, \tag{5.24}$$

hence

$$\widehat{f_2} = f_2, \qquad \widehat{f_3} = f_3. \tag{5.25}$$

These zeros are exact results of the Schur complement, not assumptions and not artifacts of the canonical (q, u, t) chart. Raw Euler–Lagrange substitution, the Schur complement, and substitution of the complete raw solution explicitly give $\Delta_2 = \Delta_3 = 0$ and the same on-shell density. Horizontal-basis independence follows algebraically from the congruence transformation of the transverse quadratic form under an invertible GL(3) change of coordinates.

5.9 Residue equation

Stationarity of (5.22) is

$$2f_2\, t^\mu - \partial^\mu f_3 = 0. \tag{5.26}$$

On $f_2 \neq 0$,

$$t_\star = \frac{\partial f_3}{2f_2}. \tag{5.27}$$

Combining (5.19) with (5.27) and writing

$$r := \frac{\partial f_1}{f_1}, \qquad s := \frac{\partial f_3}{2f_2}, \tag{5.28}$$

one obtains the complete horizontal solution

$$t = s, \qquad q = 4r - \tfrac{8}{3}s, \qquad u = r - \tfrac{2}{3}s. \tag{5.29}$$

Equivalently,

$$q_\star = 4\frac{\partial f_1}{f_1} - \frac{4}{3}\frac{\partial f_3}{f_2}, \qquad u_\star = \frac{\partial f_1}{f_1} - \frac{1}{3}\frac{\partial f_3}{f_2}. \tag{5.30}$$

No step in (5.21)–(5.29) extends through $f_1 = 0$ or $f_2 = 0$.

6 Application: kernel, orbit, and commutativity

6.1 Transverse determinant

Because $H_\perp$ is block-Schur with Palatini block P and complement $S_t = 2f_2$,

$$\det H_\perp = (\det P)\,(2f_2) = -\frac{f_1^2 f_2}{6}. \tag{6.1}$$

Therefore the species matrix $H_\perp$ is invertible if and only if $f_1 \neq 0$ and $f_2 \neq 0$. On that locus $\operatorname{rank}_{\mathrm{sp}}(H_\perp) = 3$. By (5.11) the pointwise one-form-fibre rank is then $\operatorname{rank} \mathcal{H}_\perp = D \cdot 3$, hence 12 at $D = 4$.

6.2 Full affine Hessian

Restore the vertical coordinate V. Write $S_{\mathrm{full}} \simeq \mathbb{R}^4$ for the species space of (q, u, t, V) and

$$E_{\mathrm{full},x} = S_{\mathrm{full}} \otimes T_x^* M. \tag{6.2}$$

The bulk density (5.8) is independent of V, so the species Hessian in (q, u, t, V) is the 4×4 coefficient matrix

$$H_{\mathrm{full}} = \begin{pmatrix} H_\perp & 0 \\ 0 & 0 \end{pmatrix}. \tag{6.3}$$

The true pointwise Hessian on the full one-form fibre is

$$\mathcal{H}_{\mathrm{full}} = H_{\mathrm{full}} \otimes g^{-1}. \tag{6.4}$$

Because g^{-1} is nondegenerate,

$$\ker \mathcal{H}_{\mathrm{full}} = \ker H_{\mathrm{full}} \otimes T_x^* M, \qquad \operatorname{rank} \mathcal{H}_{\mathrm{full}} = D \operatorname{rank}_{\mathrm{sp}}(H_{\mathrm{full}}). \tag{6.5}$$

On U_{reg} one has $\operatorname{rank}_{\mathrm{sp}}(H_{\mathrm{full}}) = 3$ and $\operatorname{null}_{\mathrm{sp}}(H_{\mathrm{full}}) = 1$. At $D = 4$ the corresponding fibre multiplicities are $\operatorname{rank} \mathcal{H}_{\mathrm{full}} = 12$ and $\dim \ker \mathcal{H}_{\mathrm{full}} = 4$. Species inclusion is

$$\ker H_{\mathrm{full}} \supseteq \operatorname{span}\{e_V\}, \qquad e_V = (0, 0, 0, 1). \tag{6.6}$$

Define the pointwise projective gauge map

$$G_x : T_x^* M \to E_{\mathrm{full},x}, \qquad \xi_\mu \mapsto e_V \otimes (3\xi_\mu). \tag{6.7}$$

The factor 3 is the vertical shift $\delta V = 3\xi$ of (4.1); any nonzero scalar yields the same image. Thus

$$\operatorname{im} G_x = \operatorname{span}\{e_V\} \otimes T_x^* M. \tag{6.8}$$

Proposition 6.1 (Exact kernel)**.** *On U_{reg}, the species kernel is*

$$\ker H_{\mathrm{full}} = \mathrm{span}\{e_V\}, \tag{6.9}$$

and the true pointwise kernel on the full one-form fibre is

$$\ker \mathcal{H}_{\mathrm{full}} = \mathrm{im}\, G_x. \tag{6.10}$$

Equivalently, in raw traces the species kernel is $\ker H_{\mathrm{raw}} = \mathrm{span}\{(3,-8,-2,3)\}$*, and the raw full-fibre kernel is*

$$\big\{(3,-8,-2,3)\,\xi_\mu : \xi \in T_x^*M\big\}. \tag{6.11}$$

This is a fibre-level restatement of the species result, not a new algebraic theorem. The projective gauge image exhausts the affine degeneracy of (3.7) *in the present trace sector: there is no additional Hessian kernel.*

Proof. Inclusion $\mathrm{span}\{e_V\} \subseteq \ker H_{\mathrm{full}}$ is (6.3). For the converse, if $H_{\mathrm{full}}(x,V) = 0$ with $x = (q,u,t)$, then $H_\perp x = 0$. On U_{reg}, $\det H_\perp \neq 0$, so $x = 0$, while V remains arbitrary. Thus $\ker H_{\mathrm{full}} \subseteq \mathrm{span}\{e_V\}$. Tensoring with T_x^*M and using (6.5)–(6.8) yields (6.10).

Independently, the raw bulk density after the admitted integrations by parts is quadratic in $(T, Q, \widetilde{Q}, A)$ and annihilates the projective species vector $(3,-8,-2,3)$ identically (Appendix F). On U_{reg} that species kernel is one-dimensional, so the fibre kernel is D-dimensional. The coordinate change (4.15) identifies it with $\mathrm{im}\, G_x$. The vertical Euler–Lagrange combination

$$\mathrm{EL}_T - \tfrac{8}{3}\mathrm{EL}_Q - \tfrac{2}{3}\mathrm{EL}_{\widetilde{Q}} + \mathrm{EL}_A \tag{6.12}$$

vanishes identically, as required of a gauge identity. □

Varied-compensator Noether redundancy. In raw variables the off-shell projective identity is

$$3\,\mathrm{EL}_T - 8\,\mathrm{EL}_Q - 2\,\mathrm{EL}_{\widetilde{Q}} + 3\,\mathrm{EL}_A = 0, \qquad \mathrm{EL}_A = -\mathrm{EL}_T + \frac{8}{3}\mathrm{EL}_Q + \frac{2}{3}\mathrm{EL}_{\widetilde{Q}}.$$

Thus varying A_μ introduces neither an independent bulk Euler–Lagrange constraint nor a propagating degree of freedom. It makes the complete projective auxiliary fibre explicit, so that the vertical gauge direction is represented inside the full Hessian. The model-specific result is the resulting full-fibre quotient classification and the detection of any additional invariant degeneracy, not new independent A_μ dynamics.

6.3 Complete connection solution

Translating (5.29) back to raw traces, with V arbitrary,

$$\begin{aligned} T &= s + V, \\ Q &= 4r - \tfrac{8}{3}(s+V), \\ \widetilde{Q} &= r - \tfrac{2}{3}(s+V), \\ A &= V. \end{aligned} \tag{6.13}$$

The reconstructed distortion on this solution is

$$N^\lambda{}_{\mu\nu} = \tfrac{1}{2}\big(\delta^\lambda{}_\mu r_\nu + \delta^\lambda{}_\nu r_\mu - g_{\mu\nu} r^\lambda\big) - \tfrac{1}{3}\delta^\lambda{}_\nu (s_\mu + V_\mu). \tag{6.14}$$

Shifting the representative by $V \mapsto V + 3\xi$ changes Γ by exactly $-\delta^\lambda{}_\nu \xi_\mu$, which is (4.1).

Theorem 6.2 (Local connection orbit)**.** *Assume Assumptions 3.1–4.4 and the law* (4.1)*. On* U_{reg}*, at fixed admissible* (g, ϕ)*:*

1. *there is one pointwise algebraic solution of the trace-sector affine Euler–Lagrange equations, unique up to the raw projective vector* $(3, -8, -2, 3)$*;*
2. *if the coefficient functions and the regularity conditions are smooth on a spacetime-local regular region, these pointwise solutions assemble into one spacetime-local orbit modulo the projective transformation* (4.1)*.*

The explicit solution is (6.13)–(6.14)*. Smoothness of the coefficient functions on a regular region is the regularity actually used; no Banach-manifold quotient or global slice is invoked. The statement does not classify Hamiltonian or covariant-phase-space charges of* ξ*, and is not a statement about deleted affine irreps.*

The qualifier "the connection solution" is therefore to be read as this local orbit, never as a globally unique connection.

6.4 The two routes

Let U_{reg} be as in (4.16), and let bulk densities be identified modulo the admitted class (5.7).

Route QE **(quotient first).** Replace the raw traces $(T, Q, \widetilde{Q}, A)$ by the invariant horizontal coordinates (t, q, u), discarding V. Solve the nonsingular three-species system $H_{\perp}x + J = 0$ on U_{reg}. Substitute the unique horizontal solution (5.29) into (5.8).

Route EQ **(eliminate first).** Retain an arbitrary representative V. Substitute the complete raw solution (6.13) into the original trace-expanded bulk density, before any horizontal reduction.

These are not merely names of a diagram: each is an explicit substitution, written in Appendix G. In local field theory, substitution of some fields by local functionals of others commutes with the Euler–Lagrange problem when those fields are auxiliary, and need not commute otherwise [12]. That general fact is the conceptual setting of QE = EQ. It is not a substitute for the Hessian computation of Section 6 or for the substitutions of Appendix G.

6.5 Equality of on-shell densities

Theorem 6.3 (Local bulk quotient–elimination commutativity)**.** *On* U_{reg}*, as spacetime-local bulk functionals modulo the displayed total divergences generated by* J_1 *and* J_3*,*

$$\mathcal{L}^{\mathrm{QE}}_{\mathrm{aff,on\text{-}shell}} = \mathcal{L}^{\mathrm{EQ}}_{\mathrm{aff,on\text{-}shell}} = \frac{3(\partial f_1)^2}{2f_1} - \frac{(\partial f_3)^2}{4f_2}. \tag{6.15}$$

The eliminate-first density is manifestly independent of V*. Equality is a bulk identification. It does not classify projective charges, does not extend through singular branches, and does not supply a global gauge slice.*

Proof. The substitutions are written in Appendix G. On U_{reg} the horizontal system has the unique solution (5.29). Substituting into (5.8) produces the right-hand side of (6.15) by the Schur evaluation (5.21) together with (5.27); this is QE. For EQ, substitute (6.13) into the raw quadratic-plus-linear density (F.1). Every term involving V cancels by projective invariance of that density (Proposition 6.1), and the remaining expression coincides with (6.15). The two substitutions therefore agree as bulk densities. Identification uses the admitted currents (5.7). Whether those total divergences are retained, canceled, or discarded is a choice of variational boundary data, independent of any proper/improper classification of ξ (Appendix H). □

Gauge fixing does not replace the kernel test. The local choice $V = A = 0$ removes the known vertical projective direction but does not establish invertibility of the horizontal Hessian. On U_{reg}, once $\ker \mathcal{H}_{\mathrm{full}} = \mathrm{im}\, G_x$ has been proved, gauge-fix-first, quotient-first, and eliminate-first give the same horizontal solution in the spacetime-local sense and admitted boundary class stated above. On a singular branch, fixing $V = 0$ leaves every horizontal gauge-invariant kernel direction untouched and therefore cannot remove invariant moduli or cure source incompatibility. Gauge fixing is consistent here, but the full-Hessian analysis supplies the stronger eliminability test.

The same bulk value is obtained for the representatives $V = 0$, $V = \mathrm{d}\chi$, and V an arbitrary one-form. Representative independence of the bulk is therefore part of (6.15), not an extra assumption.

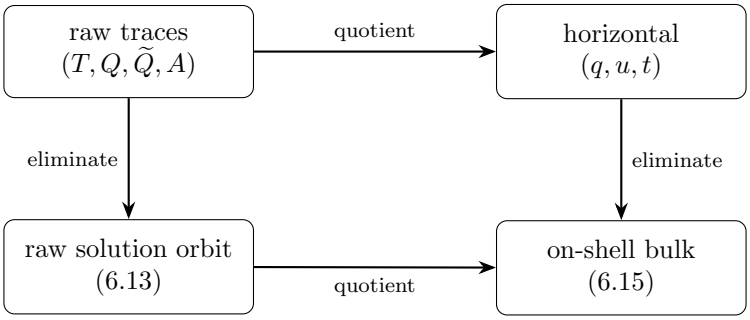


Figure 1: Commutative diagram for Theorem 6.3. Both arrows to the on-shell bulk density are explicit substitutions (Appendix G), equal as spacetime-local bulk functionals modulo the admitted IBP currents J_1, J_3, on U_{reg}.

6.6 Where the local and IBP hypotheses enter

Manifest V-independence of the eliminate-first bulk density follows from exact projective invariance, equivalently the explicit absence of V in the horizontal density. On $f_1 f_2 \neq 0$, $\ker \mathcal{H}_{\mathrm{full}} = \mathrm{im}\, G_x$ excludes additional gauge-invariant null directions and makes the horizontal quotient solve unique. Identification of the two bulk functionals uses (5.5)–(5.6) and discards $\partial_\mu(\sqrt{-g}(c_1 J_1^\mu + c_3 J_3^\mu))$ as a variational-boundary convention. Passage from a pointwise algebraic orbit to a spacetime-local orbit uses smoothness of (f_i, V) on a regular region. None of these steps is a global slice, and none classifies projective charges.

7 Reduced action

7.1 Bulk functional

Collecting $f_1 R_{\mathrm{LC}}$, the on-shell affine density (6.15), and the affine-independent potential, one obtains

$$S_{\mathrm{red}} = \int \mathrm{d}^4x\, \sqrt{-g} \left[f_1\, R_{\mathrm{LC}} + \frac{3(\partial f_1)^2}{2 f_1} - \frac{(\partial f_3)^2}{4 f_2} + U(\phi) \right] + \text{admitted boundary}. \tag{7.1}$$

The boundary addendum is of the class (5.7). Equation (7.1) is a spacetime-local bulk functional on U_{reg}, not a complete initial-boundary-value action.

7.2 Coefficient functions of a scalar

If $f_i = f_i(\phi)$ and $U = U(\phi)$, then $\partial_\mu f_i = f_i' \, \partial_\mu \phi$ and

$$K_{\mathrm{Pal}}(\phi) = \frac{3(f_1')^2}{2f_1}, \tag{7.2}$$

$$K_{\mathrm{eff}}(\phi) = \frac{3(f_1')^2}{2f_1} - \frac{(f_3')^2}{4f_2}. \tag{7.3}$$

The reduced bulk Lagrangian then reads $f_1 R_{\mathrm{LC}} + K_{\mathrm{eff}} \, (\partial\phi)^2 + U(\phi)$. The writing

$$\frac{3}{2}\frac{(\partial f_1)^2}{f_1} \tag{7.4}$$

is the same Palatini density as (7.2). The schematic ratio-square

$$\frac{3}{2}\Big(\frac{f_1'}{f_1}\Big)^2 \tag{7.5}$$

is not equivalent: it differs from (7.2) by a factor $1/f_1$ and is not used. The coefficient $3(f_1')^2/(2f_1)$ is the generic Palatini kinetic contribution associated with eliminating an independent connection in $f_1 R(\Gamma)$ [4, 7, 8]. Its presence here is reproduced structure, not a new Palatini identity.

Proposition 7.1 (Kinetic identification)**.** *The Palatini contribution generated by eliminating* (q, u) *is exactly* (7.2)*, equivalently* (7.4)*. The residue contribution generated by* (5.27) *is* $-(f_3')^2/(4f_2)$*. The auxiliary reduction determines* K_{eff} *as their sum but does not identify it with any independently supplied geometric constitutive factor.*

7.3 Conditional scalar role

If ϕ is varied as a dynamical field, the reduced scalar Euler–Lagrange equation is that of (7.1). If ϕ is a background argument of the coefficient functions, no scalar Cauchy problem is asserted. The two modes are not mixed. Auxiliary elimination does not, by itself, generate a constitutive predictivity theorem: it produces (7.1) on U_{reg}, and nothing further about the observational content of K_{eff}.

7.4 Main theorem

Theorem 7.2 (Exact local projective-affine reduction)**.** *Assume the trace/vector domain of Section 3, the projective law* (4.1)*, metric-minimal matter, and the admitted IBP class of Assumption 4.4. On* $U_{\mathrm{reg}} = \{f_1 \neq 0\} \cap \{f_2 \neq 0\}$*:*

1. *the affine Hessian kernel is exactly the projective tangent, and the transverse Hessian is invertible;*
2. *affine solutions form one local orbit in the sense of Theorem 6.2;*
3. *quotient and auxiliary elimination commute in the sense of Theorem 6.3;*
4. *the reduced bulk functional is* (7.1)*, with* (7.2)–(7.3) *when* $f_i = f_i(\phi)$*.*

The scalar role remains conditional. Signs $f_1 > 0$*,* $f_2 > 0$*, and* $K_{\mathrm{eff}} > 0$ *are not hypotheses. Deleted irreps remain outside the domain.*

8 Singular strata and limitations

8.1 Four algebraic strata

The vanishing of $\det H_\perp = -f_1^2 f_2/6$ defines two hypersurfaces in coefficient space. Together they partition the problem into four strata. The horizontal system is

$$H_\perp x + J = 0, \qquad x = (q, u, t)^{\mathsf{T}}, \tag{8.1}$$

with J as in (5.14). The classification below uses only this linear algebra: no division by f_1 or f_2 is performed until invertibility has been established.

Table 2: Regular versus singular strata of the affine Hessian, including source compatibility of (8.1). Rank and nullity columns are species-space values of the coefficient matrices. In $D = 4$ the corresponding pointwise one-form-fibre ranks and nullities are obtained by multiplication by four because $g^{\mu\nu}$ is nondegenerate. The reduced action of Theorem 7.2 exists only on S1. The vertical coordinate V remains a projective gauge parameter on every compatible stratum.

	condition	$\operatorname{rank}_{\mathrm{sp}} P$	$\operatorname{rank}_{\mathrm{sp}} H_\perp$	$\operatorname{null}_{\mathrm{sp}} H_{\mathrm{full}}$	source / existence	invariant moduli	Thm. 7.2
S1	$f_1 \neq 0,\ f_2 \neq 0$	2	3	1	always solvable; unique x	none	yes
S2	$f_1 = 0,\ f_2 \neq 0$	0	1	3	$\partial f_1 = 0$; else none	if compatible: q, u; $t = \partial f_3/(2f_2)$	no
S3	$f_1 \neq 0,\ f_2 = 0$	2	2	2	$\partial f_3 = 0$; else none	if compatible: t; q, u of t	no
S4	$f_1 = 0,\ f_2 = 0$	0	0	4	$\partial f_1 = \partial f_3 = 0$; else none	if compatible: q, u, t	no

At $D = 4$ the fibre multiplicities are therefore S1: $\operatorname{rank} \mathcal{H}_{\mathrm{full}} = 12$, $\dim \ker \mathcal{H}_{\mathrm{full}} = 4$; S2: 4 and 12; S3: 8 and 8; S4: 0 and 16. Source-compatibility statements are unchanged.

On S1 the species kernel is exactly vertical, equivalently $\ker \mathcal{H}_{\mathrm{full}} = \operatorname{im} G_x$, and the reduced action (7.1) exists. On S2, compatibility requires $\partial f_1 = 0$; if that holds, $t = \partial f_3/(2f_2)$ while q and u remain invariant algebraic moduli, whereas $\partial f_1 \neq 0$ yields no stationary horizontal solution. On S3, compatibility requires $\partial f_3 = 0$; if that holds, t is a gauge-invariant modulus and (q, u) are determined as functions of t, whereas $\partial f_3 \neq 0$ yields no stationary solution. On S4, compatibility requires $\partial f_1 = \partial f_3 = 0$; if that holds, (q, u, t) are horizontal moduli, whereas otherwise there is no stationary solution. Extra kernel on S2–S4 consists of these invariant degeneracies: it is not additional projective gauge. No additional algebraic factor of $\det H_\perp$ exists, and no regular proof step extends through S2–S4. Those branches are not solved dynamically here.

Compatible singular strata and stationary values. Direct substitution in (5.8), without extending Theorem 7.2, gives

$$\begin{aligned}
&\text{S2}: \quad \mathcal{L}_{\mathrm{aff,on}} = -(\partial f_3)^2/(4f_2), \quad q, u \text{ arbitrary},\\
&\text{S3}: \quad \mathcal{L}_{\mathrm{aff,on}} = 3(\partial f_1)^2/(2f_1), \quad t \text{ arbitrary},\\
&\text{S4}: \quad \mathcal{L}_{\mathrm{aff,on}} = 0, \quad q, u, t \text{ arbitrary}.
\end{aligned}$$

These values assume the compatibility conditions in Table 2; in S2, $t = \partial f_3/(2f_2)$, while in S3 (q, u) depend affinely on t. A unique stationary bulk value therefore does not imply a unique quotient solution, consistently with the qualification after Theorem 2.1.

Figure 2 maps the four regions.

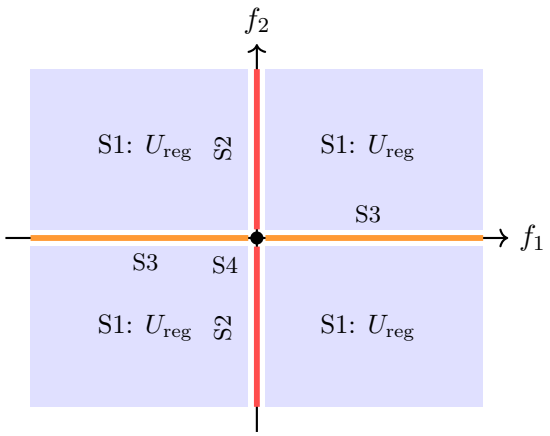


Figure 2: Regular branch S1 versus singular strata S2–S4 in the complete (f_1, f_2) coefficient plane. Algebraic regularity is $f_1 \neq 0$ and $f_2 \neq 0$: S1 comprises all four open quadrants. The vertical axis excluding the origin is S2; the horizontal axis excluding the origin is S3; the origin is S4. No sign condition is imposed by the algebraic regularity theorem. Axes themselves are excluded from Theorem 7.2.

8.2 Necessary counterexamples

The hypotheses are sharp in the following sense (Appendix I).

- If $f_1 = 0$, Palatini invertibility fails, and $\partial f_1 \neq 0$ makes (8.1) inconsistent.
- If $f_2 = 0$, residue invertibility fails, and $\partial f_3 \neq 0$ makes (8.1) inconsistent.
- If the projective law is replaced by $(a, b) = (1, 0)$, the Ricci target (4.6) fails.
- If deleted irreps are treated as solved Hessian coordinates, the domain premise fails.

8.3 Claim boundary

The following remain outside Theorem 7.2.

Table 3: Scope and excluded sectors. Excluded irreducible sectors lie outside the stated configuration space; they are neither proved to vanish nor removed by projective gauge symmetry.

Object or claim	Classification
axial torsion	excluded by truncation
tensor torsion	excluded by truncation
traceless nonmetricity	excluded by truncation
higher affine irreps	excluded by truncation
hypermomentum	not covered
independent matter connection sources	not covered
global gauge slice	not claimed
Hamiltonian / covariant-phase-space charges of ξ	not analyzed
boundary-charge algebra	not classified
extension through $f_1 = 0$ or $f_2 = 0$	not claimed
mandatory scalar dynamics	conditional only
$f_1 > 0$, $f_2 > 0$, $K_{\text{eff}} > 0$	not hypotheses
ghost-freedom / energy stability	not claimed
independent constitutive factor	not identified by the auxiliary reduction
constitutive predictivity	not claimed
unrestricted metric-affine (64-component) theorem	not claimed

In particular, algebraic regularity is $f_1 \neq 0$ and $f_2 \neq 0$. Physical-stability inequalities are a different and absent theorem. Whether the displayed IBP divergences are retained as boundary data is a variational convention, not a charge classification.

8.4 Symmetry-compatible deformation of the regularity surface

Consider the projectively invariant quadratic deformation (J.2). Here α and β are smooth, projectively inert scalar spectator coefficient functions on the local region. They are not varied as auxiliary variables, enter algebraically, and have no derivative coupling in this proposition. Pointwise Hessian statements use only their local values. On $f_1 \neq 0$, define

$$S_t(\alpha,\beta) = 2f_2 - \frac{16\alpha+4\beta}{3} + \frac{4\alpha^2-16\alpha\beta-11\beta^2}{3f_1}. \tag{8.2}$$

Proposition 8.1 (Deformed Schur regularity and source compatibility)**.** *Assume $f_1 \neq 0$ within the two-parameter family* (J.2)*.*

1. *If $S_t(\alpha,\beta) \neq 0$, then $H_\perp(\alpha,\beta)$ is invertible. The deformation introduces no V dependence, so the full species Hessian is* $\mathrm{diag}(H_\perp(\alpha,\beta),0)$ *and the true one-form-fibre kernel obeys*

$$\ker \mathcal{H}_{\mathrm{full}}(\alpha,\beta) = \mathrm{im}\, G_x.$$

   Theorem 2.1 therefore gives a unique pointwise horizontal quotient solution. If the coefficients are sufficiently smooth on an open region where $f_1 \neq 0$ and $S_t(\alpha,\beta) \neq 0$, the raw solutions assemble into one spacetime-local projective orbit. Since (J.2) *adds no derivative term and no boundary current, the same admitted IBP class gives local bulk quotient/elimination commutativity in the sense of Theorem 6.3.*

2. *If $S_t(\alpha,\beta) = 0$, the Palatini pair (q,u) remains solvable, but the residual equation is compatible only if*

$$\partial f_3 = \frac{\partial f_1}{f_1}(4\alpha+\beta).$$

   If this condition fails, there is no stationary horizontal affine solution. If it holds, t is one additional gauge-invariant algebraic modulus and (q,u) are determined as affine functions of t. The additional null direction is not projective gauge.

No statement is made here about $f_1 = 0$ in the deformed family or about unrestricted metric-affine gravity.

Remark 8.2 (A visible displacement of the rank-changing surface). Let $\beta = 0$ and $\alpha = \epsilon f_1$, where ϵ is a projectively inert constant and $f_1 \neq 0$. Then

$$S_t(\epsilon f_1, 0) = 2f_2 + \frac{4f_1}{3}\epsilon(\epsilon-4).$$

At $f_2 = 0$ the deformed horizontal system is therefore regular for $\epsilon \neq 0, 4$. This shows directly that the undeformed $f_2 = 0$ singular surface is not imposed by projective symmetry: that symmetry fixes the unavoidable vertical gauge kernel, whereas invariant coupling data determine additional horizontal degeneracy. The statement concerns a neighboring deformed model, not the undeformed action (3.7) becoming regular at the same parameter point.

Indeed, Appendix J gives $\det H_\perp(\alpha,\beta) = -(f_1^2/12)S_t(\alpha,\beta)$ and the displayed effective source. The undeformed singular surface $f_2 = 0$ is therefore not universal under nearby projectively invariant quadratic mixing: within this family the invariant rank-changing condition is the Schur surface $S_t(\alpha,\beta) = 0$. The distinction between the projective gauge kernel and an additional invariant auxiliary kernel persists, while the locus at which the latter appears moves with the invariant couplings.

9 Discussion

9.1 Quotient geometry and auxiliary nonuniqueness

Raw traces $(T, Q, \widetilde{Q}, A)$ split as $(q, u, t \mid V)$. The vertical coordinate $V_\mu = A_\mu$ parametrizes the projective orbit, while (q, u, t) provide invariant horizontal affine data. These horizontal variables are not gauge-removable, yet on U_{reg} they are algebraic. In particular $t = T - A$ is projectively invariant and is solved by $t_\star = \partial f_3/(2f_2)$. Thus gauge invariance does not imply propagation. After elimination, $f_i(\phi)$ and $U(\phi)$ remain constitutive data: auxiliary linear algebra fixes K_{eff} but does not identify it with an independently supplied geometric constitutive factor.

The logical hierarchy is the following. Theorem 2.1 is the general finite-dimensional organizing criterion; evaluating the full one-form-fibre kernel is the model-specific geometric work. The unique local orbit is a consequence of the exact equality $\ker \mathcal{H}_{\mathrm{full}} = \operatorname{im} G_x$, while Theorem 6.3 is an explicit consistency check of the quotient-first and eliminate-first routes. The singular source conditions add information precisely where rank is lost, and Proposition 8.1 tests the transferability of this kernel/source mechanism in a nearby symmetry-compatible family. These statements are therefore logically related rather than independent general breakthroughs. Unrestricted necessity of commutativity is not claimed: on-shell *values* can agree for a special compatible source even if extra kernel is present.

9.2 Fingerprint $\Delta_2 = \Delta_3 = 0$

Palatini elimination does not renormalize the residue couplings: $\Delta_2 = \Delta_3 = 0$. Appendix J tests that fingerprint against the two-parameter quadratic deformation family (J.2), with (3.7) as its undeformed origin. Requiring both corrections to vanish as identities over the admitted coefficient-function class selects the undeformed point. This structural statement does not exclude accidental cancellations on a fixed special background. An independent linear-source deformation is a separate supporting control and is not part of the proposition below. Thus $\Delta_2 = \Delta_3 = 0$ is a structural fingerprint of (3.7) within the tested family, not a claim about all conceivable symmetry-compatible operators.

Proposition 9.1 (Structural non-genericity of the residue fingerprint)**.** *Within the two-parameter family* (J.2)*, require the corrections to the residue Schur coefficient and effective residue source to vanish as algebraic identities over the admitted coefficient-function class with* $f_1 \neq 0$*. Both structural corrections vanish if and only if* $(\alpha, \beta) = (0, 0)$*.*

Proof. Multiplying (J.5) by $3f_1$ shows that identity in f_1 requires $4\alpha + \beta = 0$. Structural source preservation in (J.6) gives the same condition for arbitrary ∂f_1. Substitution of $\beta = -4\alpha$ into the f_1-independent remainder gives $-108\alpha^2$, hence $\alpha = \beta = 0$. The converse is immediate. □

Remark 9.2 (Accidental fixed-background cancellations)*.* The proposition is not a pointwise uniqueness statement. For example, on a fixed background with $f_1 = c \neq 0$ and $\partial f_1 = 0$, the nonzero deformation $(\alpha, \beta) = (0, -4c/11)$ makes both corrections vanish.

9.3 Singular crossings of $f_i(\phi)$

If $f_i = f_i(\phi)$, a locus $f_1(\phi_*) = 0$ has $\partial f_1 = f_1'(\phi_*)\,\mathrm{d}\phi$. A generic crossing with $f_1'(\phi_*) \neq 0$ and $\mathrm{d}\phi \neq 0$ therefore violates the S2 compatibility condition $\partial f_1 = 0$ and yields no stationary affine solution. Only a source-compatible crossing can support the S2 moduli branch. On $f_2(\phi_*) = 0$, compatibility is controlled by $\partial f_3 = 0$, not by ∂f_2. This is an algebraic reading of Table 2; it is not a claim of dynamical singularity propagation or of cosmological pathology.

9.4 Relation to metric-affine literature

Projective Palatini gravity and Ricci-based MAG connection solutions are known [2, 3, 6, 9, 22]. Palatini $f(R)$ and the kinetic shift $3(F')^2/(2F)$ are known [4, 5, 7, 8]. Projectively invariant scalar-metric-affine models integrate the connection out to DHOST-type theories [10, 23]. Quadratic MAG constructions often solve the connection after breaking or otherwise controlling projective degeneracy [11, 13]. Ferrara, Guzmán, and Järv already treat singular connection operators, symmetry-generated versus accidental zero modes, source constraints, and symmetry breaking in broader quadratic MAG [14]; none of those general structures is a priority claim here. The narrower contribution is their closed realization in the specified compensated trace/vector system: exact full-fibre kernel equality, quotient eliminability, local orbit, explicit QE/EQ bulk comparison, undeformed singular-source classification, and the controlled shifted-Schur extension. Unified Palatini treatments with torsion or nonmetricity give precise conditions for propagating versus algebraic distortion [24]. Auxiliary substitution commutativity [12] is the conceptual setting, not a substitute MAG proof. Table 4 records a positive-scope comparison of these sources.

Recent scalar-metric-affine work likewise eliminates an algebraic independent connection on a generic nondegenerate branch and encodes its response in an effective scalar kinetic function [25]. This application motivates the complementary structural questions here: when is such elimination unique modulo projective gauge, and what source compatibility remains when the auxiliary rank changes?

The condition $\ker H = \operatorname{im} G$ is the general finite-dimensional auxiliary diagnostic used here. By contrast, $\det H_\perp = -f_1^2 f_2/6$ and the exact realization of the projective gauge kernel hold for the stated trace/vector configuration space. Restoring omitted affine irreducible sectors would require a new Hessian analysis; this sector is not established as a dynamically consistent truncation of unrestricted 64-component MAG [1, 18].

Table 4: Positive-scope comparison of related metric-affine and substitution sources. Entries record established content in each work's own setting. Related residue and reduction-order sources are discussed in the Introduction.

Work	Affine setting	Principal reduction/connection result	Relation to the present question
Dadhich–Pons [3]	Palatini Einstein–Hilbert	Projective orbit of the Palatini connection; Levi-Civita plus projective mode	Projective equivalence of Palatini gravity; independent of the present Hessian test
Alfonso et al. [9]	Ricci-based MAG	Algebraic connection solution with projective invariance retained; torsion role trivialized in the observables	Ricci-based algebraic Γ solve; distinct configuration space
Aoki–Shimada [10, 23]	Scalar metric-affine	Connection integrated out to a projectively protected scalar-metric theory	Projective ghost protection after Γ elimination; different operator content
Iosifidis 2019 [11]	Solvable MAG	Explicit algebraic connection solutions under stated solvability criteria	Algebraic Γ solvability in a broader MAG class
Iosifidis 2022 [13]	Quadratic MAG	Quadratic connection solve under nondegeneracy hypotheses, often with controlled projective data	Unique representative after nondegeneracy hypotheses
Ferrara–Guzmán–Järv 2026 [14]	Generalized-projective quadratic MAG	Singular operators, symmetry-generated and accidental zero modes, source constraints, and symmetry breaking	General framework; the present work gives the closed classification for the specified varied-A_μ system
Andrei–Iosifidis–Järv–Saal 2026 [25]	Scalar metric-affine	Algebraic connection eliminated on a generic nondegenerate branch, yielding an effective scalar kinetic sector	Uniqueness modulo projective gauge and source compatibility when rank changes
Böhmer–Jensko [24]	Unified Palatini MAG	Criteria distinguishing algebraic versus propagating distortion	Distortion-propagation tests in Palatini MAG
Pons 2010 [12]	Local field substitution	Conditions under which substituting fields in the action commutes with the Euler–Lagrange problem	Conceptual setting of QE = EQ; not a MAG model
present work	Trace/vector residue $T - A$	Exact projective kernel on the full one-form fibre; unique local orbit and QE = EQ on U_{reg}	Species determinant, source-compatible singular strata, and shifted Schur surface in a two-parameter family

9.5 Scope of the reduction

Solving Γ and A produces (7.1). The result is confined to the trace/vector configuration space, is spacetime local, assumes metric-minimal matter, and uses the admitted boundary class. It does not establish a full-MAG embedding, stability, charge equivalence, or a global projective slice [2].

10 Conclusion

Gauge nonuniqueness and auxiliary nonuniqueness need not coincide. For a local quadratic auxiliary density, invariance along a gauge distribution together with $\ker H = \operatorname{im} G$ is the finite-dimensional criterion separating pure gauge degeneracy from additional invariant fibre degeneracy. In the metric-affine trace/vector model (3.7), the curvature and compensator targets select (4.1) within the adopted two-parameter family. On $f_1 \neq 0$ and $f_2 \neq 0$ the species determinant is $\det H_\perp = -f_1^2 f_2/6$, so the pointwise one-form-fibre kernel is exactly the projective gauge image. The quotient solution is unique, affine solutions form one local orbit, QE = EQ holds as spacetime-local bulk functionals modulo the displayed currents J_1 and J_3, and the reduced functional is (7.1) with the standard Palatini kinetic term $3(f_1')^2/(2f_1)$.

On S2 ($f_1 = 0$, $f_2 \neq 0$), stationary affine solutions require $\partial f_1 = 0$; on S3 ($f_1 \neq 0$, $f_2 = 0$), they require $\partial f_3 = 0$; and on S4 ($f_1 = f_2 = 0$), both conditions are required. Rank loss can thus mean source incompatibility or invariant algebraic moduli, rather than an enlarged projective orbit. The model thereby gives an exact local realization of the kernel criterion, including its singular-stratum geometry. In the two-parameter symmetry-compatible deformation family, the same distinction persists while the regularity boundary moves to the explicit surface $S_t(\alpha, \beta) = 0$, with a corresponding closed-form source condition. These conclusions remain local to the stated auxiliary sector and do not extend the regular theorem through rank-changing strata.

A Conventions and projective transformation

This appendix reconstructs Table 1 and Proposition 4.1.

A.1 Index conventions

$D = 4$, signature $(-,+,+,+)$. $A_{(\mu\nu)} = (A_{\mu\nu} + A_{\nu\mu})/2$ and $A_{[\mu\nu]} = (A_{\mu\nu} - A_{\nu\mu})/2$. The curvature, Ricci, torsion, and nonmetricity definitions are (3.1)–(3.5).

A.2 Trace map of the two-parameter family

Let $\delta\Gamma^\lambda{}_{\mu\nu} = a\delta^\lambda{}_\mu\xi_\nu + b\delta^\lambda{}_\nu\xi_\mu$ and $\delta g_{\mu\nu} = 0$. Then

$$\delta T^\lambda{}_{\mu\nu} = (a-b)\big(\delta^\lambda{}_\mu\xi_\nu - \delta^\lambda{}_\nu\xi_\mu\big), \tag{A.1}$$

hence

$$\delta T_\mu = \delta T^\lambda{}_{\lambda\mu} = (a-b)\big(D\xi_\mu - \xi_\mu\big) = (D-1)(a-b)\xi_\mu. \tag{A.2}$$

At $D = 4$ this is $\delta T = 3(a-b)\xi$. For nonmetricity, $Q_{\lambda\mu\nu} = -\nabla_\lambda g_{\mu\nu}$ and $\delta g = 0$ give

$$\delta Q_{\lambda\mu\nu} = \delta\Gamma^\rho{}_{\lambda\mu} g_{\rho\nu} + \delta\Gamma^\rho{}_{\lambda\nu} g_{\mu\rho}. \tag{A.3}$$

With $\delta\Gamma^\rho{}_{\lambda\mu} = a\delta^\rho{}_\lambda\xi_\mu + b\delta^\rho{}_\mu\xi_\lambda$,

$$\delta Q_{\lambda\mu\nu} = a\big(g_{\lambda\nu}\xi_\mu + g_{\lambda\mu}\xi_\nu\big) + 2b\, g_{\mu\nu}\xi_\lambda. \tag{A.4}$$

Contracting,

$$\delta Q_\lambda = g^{\mu\nu}\delta Q_{\lambda\mu\nu} = (2a + 2Db)\xi_\lambda, \tag{A.5}$$

$$\delta\widetilde{Q}_\nu = g^{\mu\lambda}\delta Q_{\lambda\mu\nu} = \big((D+1)a + 2b\big)\xi_\nu. \tag{A.6}$$

At $D = 4$ this is (4.4):

$$\delta Q = (2a + 8b)\xi, \qquad \delta\widetilde{Q} = (5a + 2b)\xi. \tag{A.7}$$

A.3 Linear Ricci coefficients

In a local inertial frame the partial terms of (3.1) with $C^\lambda{}_{\mu\nu} = a\delta^\lambda{}_\mu\xi_\nu + b\delta^\lambda{}_\nu\xi_\mu$ give, after Ricci contraction $R_{\alpha\beta} = R^\lambda{}_{\alpha\lambda\beta}$,

$$\delta R_{\alpha\beta}\big|_\partial = b\,\partial_\alpha\xi_\beta + (-3a - b)\,\partial_\beta\xi_\alpha. \tag{A.8}$$

The quadratic remainder $C{\cdot}C$ in the connection-difference identity is proportional to a in this family and vanishes identically at $a = 0$. The target $\delta R_{\mu\nu} = -2\partial_{[\mu}\xi_{\nu]}$ requires $b = -1$ and $-3a - b = 1$. Combined with $a - b = 1$ from $\delta T = +3\xi$ and with $\delta A = +3\xi$ imposed, the unique solution is $a = 0$, $b = -1$.

A.4 Direct curvature variation at $a = 0$, $b = -1$

Set $C^\lambda{}_{\mu\nu} = -\delta^\lambda{}_\nu\xi_\mu$. Mixed Γ–C terms cancel pairwise and $C{\cdot}C$ cancels by symmetry of $\xi_\mu\xi_\nu$. The remainder is

$$\delta R^\lambda{}_{\rho\mu\nu} = \partial_\mu C^\lambda{}_{\nu\rho} - \partial_\nu C^\lambda{}_{\mu\rho} = -\delta^\lambda{}_\rho(\partial_\mu\xi_\nu - \partial_\nu\xi_\mu), \tag{A.9}$$

hence $\delta R_{\rho\nu} = -2\partial_{[\rho}\xi_{\nu]}$ and $\delta R = 0$, which is (4.10).

A.5 Alternative first-index shift under the present targets

The pair $(a, b) = (1, 0)$ yields $\delta T = +3\xi$ but $\delta(Q, \widetilde{Q}) = (2, 5)\xi$ and $b = 0$ in (4.5), so the Ricci target fails. It is a standard first-kind MAG writing elsewhere. Under the present targets it is not selected by the two-parameter family; it is not declared incorrect as a general convention.

B Trace-distortion inversion

B.1 One-form ansatz

On the trace-only sector, a metric-covariant tensor linear in a one-form v_μ has the form

$$N^\lambda{}_{\mu\nu}[v] = a\,\delta^\lambda{}_\mu v_\nu + b\,\delta^\lambda{}_\nu v_\mu + c\,g_{\mu\nu}v^\lambda. \tag{B.1}$$

An independent triple (a, b, c) is used for each of T, Q, and $\widetilde{Q}$.

B.2 Trace map and inverse

Let (t, q, r) be the reconstructed $(T, Q, \widetilde{Q})$ coefficients produced by one such v. Direct contraction gives

$$t = (D-1)(a-b), \qquad q = 2(a + Db + c), \qquad r = (D+1)(a+c) + 2b. \tag{B.2}$$

With $L = (D-1)(D+2) = D^2 + D - 2$, inversion yields

$$\begin{aligned} a &= \frac{t}{D-1} + \frac{(D+1)q}{2L} - \frac{r}{L}, \\ b &= \frac{(D+1)q}{2L} - \frac{r}{L}, \\ c &= -\frac{t}{D-1} - \frac{(D+3)q}{2L} + \frac{(D+1)r}{L}. \end{aligned} \tag{B.3}$$

B.3 Four-dimensional distortion

At $D = 4$, applying the inverse separately to unit T, Q, and $\widetilde{Q}$ data produces (4.13). No other irrep enters.

B.4 Horizontal change of basis

In the ordered raw basis $(T, Q, \widetilde{Q}, A)$ and horizontal basis (t, q, u, V),

$$\begin{pmatrix} t \\ q \\ u \\ V \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 8/3 \\ 0 & 0 & 1 & 2/3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} T \\ Q \\ \widetilde{Q} \\ A \end{pmatrix}. \tag{B.4}$$

The determinant is 1. The inverse is (4.15). The raw projective vector $(3, -8, -2, 3)$ is sent to $(0, 0, 0, 3)$. Any $G \in \mathrm{GL}(3)$ acting on (q, u, t) multiplies $\det H_\perp$ by $(\det G)^2$ and leaves the regular locus $f_1 f_2 = 0$ unchanged.

C Full curvature expansion

C.1 Connection-difference identity

With $\Gamma = \{\} + N$, the scalar curvature splits into R_{LC}, a divergence built from traces of N, and a quadratic form in N. In components, writing N for N,

$$R - R_{\mathrm{LC}} = \bar{\nabla}_\mu \big(N^\mu{}_\lambda{}^\lambda - N^\lambda{}_\lambda{}^\mu \big) + g^{\rho\nu} \big(N^\lambda{}_{\lambda\kappa} N^\kappa{}_{\nu\rho} - N^\lambda{}_{\nu\kappa} N^\kappa{}_{\lambda\rho} \big), \tag{C.1}$$

up to the standard rearrangement that identifies the displayed divergence with $\bar{\nabla}_\mu W^\mu$ after the trace-only substitution. (The identity is the contraction of (3.1) and does not use the projective law.)

C.2 Substitution of the trace distortion

Insert (4.13) and contract at $D = 4$. The divergence coefficients are

$$N^\mu{}_\lambda{}^\lambda - N^\lambda{}_\lambda{}^\mu = -2T^\mu - Q^\mu + \widetilde{Q}^\mu = W^\mu. \tag{C.2}$$

The quadratic polynomial is bilinear in $(T, Q, \widetilde{Q})$. Collecting T^2, Q^2, $\widetilde{Q}^2$, $T \cdot Q$, $T \cdot \widetilde{Q}$, and $Q \cdot \widetilde{Q}$ produces exactly the six coefficients

$$\big(-\tfrac{2}{3},\ -\tfrac{11}{72},\ +\tfrac{1}{18},\ -\tfrac{2}{3},\ +\tfrac{2}{3},\ +\tfrac{2}{9} \big), \tag{C.3}$$

which is (5.1). There is no seventh trace structure and no remainder in this sector.

C.3 Horizontal independence of V

Substitute (4.15). Then

$$W = -2(t + V) - \big(q - \tfrac{8}{3} V \big) + \big(u - \tfrac{2}{3} V \big) = -2t - q + u, \tag{C.4}$$

$$T^2 = (t + V)^2, \quad Q^2 = \big(q - \tfrac{8}{3} V \big)^2, \quad \widetilde{Q}^2 = \big(u - \tfrac{2}{3} V \big)^2, \tag{C.5}$$

and likewise for the three cross terms. Expanding and using the six coefficients, every monomial containing V cancels. The quadratic form in (t, q, u) is identical to that in $(T, Q, \widetilde{Q})$. This is the second, trace-polynomial, proof of $\delta R = 0$.

C.4 Hessian H_0

Let $\mathsf{Q}(q,u,t)$ denote the quadratic polynomial of (5.1) with $(T,Q,\widetilde{Q})\mapsto(q,u,t)$. The Hessian matrix with respect to (q,u,t), treating metric contractions as the inner product, is

$$H_0=\begin{pmatrix}\partial_q\partial_q\mathsf{Q} & \partial_q\partial_u\mathsf{Q} & \partial_q\partial_t\mathsf{Q}\\ \partial_u\partial_q\mathsf{Q} & \partial_u\partial_u\mathsf{Q} & \partial_u\partial_t\mathsf{Q}\\ \partial_t\partial_q\mathsf{Q} & \partial_t\partial_u\mathsf{Q} & \partial_t\partial_t\mathsf{Q}\end{pmatrix}=\begin{pmatrix}-11/36 & 2/9 & -2/3\\ 2/9 & 1/9 & 2/3\\ -2/3 & 2/3 & -4/3\end{pmatrix}, \tag{C.6}$$

because $\partial_q\partial_q(-11/72\,q^2)=-11/36$, $\partial_u\partial_u(1/18\,u^2)=1/9$, $\partial_t\partial_t(-2/3\,t^2)=-4/3$, $\partial_q\partial_u(2/9\,q\cdot u)=2/9$, $\partial_q\partial_t(-2/3\,t\cdot q)=-2/3$, and $\partial_u\partial_t(2/3\,t\cdot u)=2/3$. This is (5.13).

D Palatini block, determinant, and inverse

D.1 Source vector

After (5.5)–(5.6) the linear bulk terms are

$$-(\partial f_1)\cdot W-(\partial f_3)\cdot t=(\partial f_1)\cdot(2t+q-u)-(\partial f_3)\cdot t, \tag{D.1}$$

hence

$$J=\big(\partial f_1,\ -\partial f_1,\ 2\partial f_1-\partial f_3\big)^{\mathsf{T}}, \tag{D.2}$$

which is (5.14). The f_2t^2 term contributes $\mathrm{diag}(0,0,2f_2)$ to the Hessian in the $\frac12 x^{\mathsf{T}}Hx$ normalization.

D.2 Determinant of P

From (5.15),

$$\det\begin{pmatrix}-11/36 & 2/9\\ 2/9 & 1/9\end{pmatrix}=\big(-\tfrac{11}{36}\big)\big(\tfrac19\big)-\big(\tfrac29\big)^2=-\tfrac{11}{324}-\tfrac{4}{81}=-\tfrac{11}{324}-\tfrac{16}{324}=-\tfrac{1}{12}. \tag{D.3}$$

Thus $\det P=-f_1^2/12$, which is (5.16).

D.3 Inverse

The adjugate of the numerical matrix is

$$\begin{pmatrix}1/9 & -2/9\\ -2/9 & -11/36\end{pmatrix}\Big/(-1/12)=\begin{pmatrix}-4/3 & 8/3\\ 8/3 & 11/3\end{pmatrix}, \tag{D.4}$$

and scaling by f_1^{-1} yields (5.17). Direct multiplication confirms $PP^{-1}=I_2$ on $f_1\neq0$.

D.4 q/u solve at fixed t

Equation (5.18) is $Pq_N=-f_1c\,t-J_N$ with $c=(-2/3,2/3)^{\mathsf{T}}$ and $J_N=(\partial f_1,-\partial f_1)^{\mathsf{T}}$. Then

$$-P^{-1}J_N=\big(4r,\ r\big), \tag{D.5}$$

$$-P^{-1}(f_1c\,t)=\big(-\tfrac83 t,\ -\tfrac23 t\big), \tag{D.6}$$

with $r=\partial f_1/f_1$, which is (5.19).

E Schur calculation and complete affine solution

E.1 Schur complement

Partition $H_\perp$ as

$$H_\perp = \begin{pmatrix} P & B \\ B^{\mathsf{T}} & C \end{pmatrix}, \qquad B = f_1 \begin{pmatrix} -2/3 \\ 2/3 \end{pmatrix}, \qquad C = f_1(-4/3) + 2f_2. \tag{E.1}$$

The Schur complement in the t slot is $S_t = C - B^{\mathsf{T}} P^{-1} B$. Substituting (5.17) gives $B^{\mathsf{T}} P^{-1} B = f_1(-4/3)$, hence $S_t = 2f_2$. The effective t-source after eliminating q_N is

$$J_{t,\mathrm{eff}} = J_t - B^{\mathsf{T}} P^{-1} J_N = (2\partial f_1 - \partial f_3) - 2\partial f_1 = -\partial f_3. \tag{E.2}$$

The t-independent remainder is

$$\mathcal{L}_{\mathrm{const}} = -\tfrac{1}{2} J_N^{\mathsf{T}} P^{-1} J_N = \frac{3(\partial f_1)^2}{2f_1}. \tag{E.3}$$

This is (5.21). Consequently $\Delta_2 = \Delta_3 = 0$ and $\widehat{f_2} = f_2$, $\widehat{f_3} = f_3$.

E.2 Residue solution and reconstruction

$S_t t + J_{t,\mathrm{eff}} = 0$ is $2f_2 t - \partial f_3 = 0$, hence (5.27) on $f_2 \neq 0$. Combined with (5.19) this is (5.29). The raw reconstruction (6.13) follows from (4.15).

E.3 On-shell distortion

Insert $T = s + V$, $Q = 4r - (8/3)(s+V)$, $\widetilde{Q} = r - (2/3)(s+V)$ into (4.13). All terms rearrange into

$$N^\lambda{}_{\mu\nu} = \tfrac{1}{2}\big(\delta^\lambda{}_\mu r_\nu + \delta^\lambda{}_\nu r_\mu - g_{\mu\nu} r^\lambda\big) - \tfrac{1}{3}\delta^\lambda{}_\nu (s_\mu + V_\mu), \tag{E.4}$$

which is (6.14). The V piece is a pure last-index shift: $V \mapsto V + 3\xi$ produces $\delta N^\lambda{}_{\mu\nu} = -\delta^\lambda{}_\nu \xi_\mu$.

F Full Hessian kernel

F.1 Raw bulk density

After the admitted integrations by parts, the affine bulk density in raw traces is

$$\mathcal{L} = f_1\, \mathsf{Q}(T, Q, \widetilde{Q}) + (\partial f_1)\cdot(2T + Q - \widetilde{Q}) + f_2(T - A)^2 - (\partial f_3)\cdot(T - A), \tag{F.1}$$

with Q the quadratic polynomial of (5.1). Let $X = (T, Q, \widetilde{Q}, A)$ and let H_{raw} be the Hessian of (F.1) with respect to X.

F.2 Gauge inclusion

Differentiating (F.1) along $(3, -8, -2, 3)$ yields zero identically, because that vector is the projective orbit of Lemma 4.2 and Q together with $W = -2T - Q + \widetilde{Q}$ are invariant, while $T - A$ is invariant. Thus

$$\mathrm{span}\{(3, -8, -2, 3)\} \subseteq \ker H_{\mathrm{raw}}. \tag{F.2}$$

F.3 Converse on the regular locus

In the invariant coordinates, $\mathcal{L}$ is independent of V and equals (5.8). Therefore the species matrix is $H_{\mathrm{full}} = \mathrm{diag}(H_\perp, 0)$ and $\det H_\perp = -f_1^2 f_2/6$. On U_{reg}, $H_\perp$ is invertible, so $\ker H_{\mathrm{full}} = \mathrm{span}\{e_V\}$ and $\ker \mathcal{H}_{\mathrm{full}} = \mathrm{im}\, G_x$, which pulls back to the raw fibre kernel (6.11). This is the converse: no additional affine kernel exists in the present trace sector.

F.4 Gauge identity

The Noether identity associated with (4.1) is the vanishing of (6.12). It holds off-shell as an identity among the raw Euler–Lagrange expressions of (F.1), and is equivalent to V-independence of $\mathcal{L}$.

F.5 Singular enlargement

On S2–S4, $\mathrm{rank}_{\mathrm{sp}}(H_{\mathrm{full}})$ drops as in Table 2, so the kernel properly contains the projective tangent. Those extra null directions are not gauge: they are gauge-invariant degeneracies of the quadratic form at vanishing f_1 or f_2, matching the invariant moduli of Appendix I. Selecting a representative among them is a non-gauge restriction. They are excluded from Theorem 7.2 rather than quotiented.

G Explicit QE and EQ substitutions

G.1 Quotient first

On U_{reg}, $H_\perp x + J = 0$ has the unique solution (5.29). Then

$$\mathcal{L}^{\mathrm{QE}}_{\mathrm{aff,on\text{-}shell}} = \tfrac{1}{2} x_\star^{\mathsf{T}} H_\perp x_\star + J^{\mathsf{T}} x_\star. \tag{G.1}$$

Using the Schur factorization,

$$\tfrac{1}{2} x_\star^{\mathsf{T}} H_\perp x_\star + J^{\mathsf{T}} x_\star = \mathcal{L}_{\mathrm{const}} + \tfrac{1}{2} S_t t_\star^2 + J_{t,\mathrm{eff}} \cdot t_\star = \frac{3(\partial f_1)^2}{2 f_1} + f_2 s^2 - (\partial f_3) \cdot s, \tag{G.2}$$

with $s = \partial f_3/(2 f_2)$. The last two terms are

$$f_2 \Big(\frac{\partial f_3}{2 f_2}\Big)^2 - (\partial f_3) \cdot \frac{\partial f_3}{2 f_2} = \frac{(\partial f_3)^2}{4 f_2} - \frac{(\partial f_3)^2}{2 f_2} = -\frac{(\partial f_3)^2}{4 f_2}. \tag{G.3}$$

Hence

$$\mathcal{L}^{\mathrm{QE}}_{\mathrm{aff,on\text{-}shell}} = \frac{3(\partial f_1)^2}{2 f_1} - \frac{(\partial f_3)^2}{4 f_2}. \tag{G.4}$$

G.2 Eliminate first

Substitute (6.13) into (F.1). Write $T = s + V$, $Q = 4r - (8/3)(s + V)$, $\widetilde{Q} = r - (2/3)(s + V)$, $A = V$. Every occurrence of V in $\mathsf{Q}(T, Q, \widetilde{Q})$ and in $2T + Q - \widetilde{Q}$ cancels by Appendix C, and $T - A = s$ is independent of V. The remaining expression is a function of (r, s) only. Expanding,

$$\begin{aligned} f_1\, \mathsf{Q}\big|_{V\text{-free}} + (\partial f_1) \cdot (2s + 4r - r) + f_2 s^2 - (\partial f_3) \cdot s \\ = \frac{3(\partial f_1)^2}{2 f_1} - \frac{(\partial f_3)^2}{4 f_2}, \end{aligned} \tag{G.5}$$

after using $r = \partial f_1/f_1$, $s = \partial f_3/(2 f_2)$, and the Palatini evaluation of Q on $(q, u) = (4r - 8s/3,\, r - 2s/3)$. Thus

$$\mathcal{L}^{\mathrm{EQ}}_{\mathrm{aff,on\text{-}shell}} = \frac{3(\partial f_1)^2}{2 f_1} - \frac{(\partial f_3)^2}{4 f_2}, \tag{G.6}$$

independent of V. The representatives $V = 0$, $V = \mathrm{d}\chi$, and V arbitrary therefore yield the same bulk density.

G.3 Admitted remainder

The two substitutions are performed on the bulk densities after (5.5)–(5.6). They therefore agree modulo

$$\partial_\mu\big(\sqrt{-g}\,(c_1 J_1^\mu + c_3 J_3^\mu)\big). \tag{G.7}$$

Whether this remainder is retained, canceled, or discarded is a variational-boundary convention. Off shell, $\delta J_1 = \delta J_3 = 0$ for projectively inert spectators, so the remainder does not classify proper versus improper transformations (Appendix H).

H Boundary currents and IBP identification

H.1 Currents actually used

The integrations by parts in the proof are only (5.5) and (5.6). They generate

$$J_1^\mu = f_1 W^\mu, \qquad J_3^\mu = f_3 \bar{T}^\mu, \tag{H.1}$$

with $W^\mu = -2T^\mu - Q^\mu + \widetilde{Q}^\mu$ and $\bar{T}^\mu = t^\mu = T^\mu - A^\mu$. No other surface current is invoked.

H.2 Admitted identification

Two bulk functionals are identified when their difference is a total divergence of the linear span of J_1 and J_3. This is a definition of the admitted variational-boundary class, not a claim that all metric-affine boundary terms lie in that span, and not a classification of gauge charges.

H.3 Off-shell projective invariance of the currents

Under the adopted law (4.1),

$$\delta(T, Q, \widetilde{Q}, A) = (3, -8, -2, 3)\xi \tag{H.2}$$

holds off shell. Therefore

$$\delta t = \delta T - \delta A = 0, \tag{H.3}$$

$$\delta W = -2\delta T - \delta Q + \delta\widetilde{Q} = (-6 + 8 - 2)\xi = 0. \tag{H.4}$$

Equivalently, the inverse chart (4.15) gives $W = -2t - q + u$ identically, with no V remainder. If the spectator coefficients f_1 and f_3 are projectively inert,

$$\delta J_1^\mu = 0, \qquad \delta J_3^\mu = 0 \tag{H.5}$$

off shell. Consequently

$$n_\mu \delta(J_1^\mu + J_3^\mu) = 0 \tag{H.6}$$

holds for every projective transformation of this class. That identity cannot distinguish proper from improper projective transformations.

Two distinct issues must therefore be kept separate.

A. IBP / variational boundary treatment. Whether the displayed total divergence of J_1 and J_3 is retained, canceled, or discarded is a choice of boundary data for the variational problem. Theorem 6.3 identifies QE and EQ as spacetime-local bulk functionals modulo that divergence.

B. Gauge charges / improper transformations. A Hamiltonian or covariant-phase-space analysis of charges carried by ξ is a separate question. It is not established here, and is not attempted.

H.4 Tests

Test	Content	Inside Theorem 7.2
A	bulk QE = EQ modulo J_1, J_3	yes
B	retain, cancel, or discard the displayed IBP divergence	boundary data, not a hypothesis
C	Hamiltonian or covariant-phase-space charge of ξ	no

No global boundary classification, charge algebra, or topological sector is claimed. The reduced functional (7.1) is not asserted to be a complete initial-boundary-value action without further boundary data.

H.5 Local versus global vocabulary

Claim	Scope
Palatini inverse	field-space local, on U_{reg}
fibre kernel equals $\operatorname{im} G_x$	field-space local, on U_{reg}
connection solve	pointwise algebraic
orbit uniqueness	field-space local and spacetime local, modulo (4.1)
QE = EQ	spacetime-local bulk functionals modulo J_1, J_3
reduced functional	spacetime-local bulk plus admitted IBP remainder
global gauge slice	not claimed
projective charges of ξ	not claimed
all gauges / all connections	not claimed

I Singular strata and counterexamples

The following examples show that the hypotheses of Theorem 7.2 are necessary. They are not solutions of the singular dynamics. All statements start from $H_\perp x + J = 0$ with $x = (q, u, t)$ and $J = (\partial f_1, -\partial f_1, 2\partial f_1 - \partial f_3)^{\mathsf{T}}$, without dividing by f_1 or f_2.

S1, $f_1 \neq 0$, $f_2 \neq 0$. $\operatorname{rank}_{\mathrm{sp}}(P) = 2$, $\operatorname{rank}_{\mathrm{sp}}(H_\perp) = 3$, $\operatorname{rank}_{\mathrm{sp}}(H_{\mathrm{full}}) = 3$, $\operatorname{null}_{\mathrm{sp}}(H_{\mathrm{full}}) = 1$. At $D = 4$ the fibre values are $\operatorname{rank} \mathcal{H}_{\mathrm{full}} = 12$ and $\dim \ker \mathcal{H}_{\mathrm{full}} = 4$. The source J always lies in the image, and the unique horizontal solution is (5.29). The regular-locus theorem is unchanged.

S2, $f_1 = 0$, $f_2 \neq 0$. $P = 0$, $\operatorname{rank}_{\mathrm{sp}}(H_\perp) = 1$, $\operatorname{rank}_{\mathrm{sp}}(H_{\mathrm{full}}) = 1$, $\operatorname{null}_{\mathrm{sp}}(H_{\mathrm{full}}) = 3$. At $D = 4$ the fibre values are rank 4 and nullity 12. The system collapses to $\partial f_1 = 0$ together with $2 f_2\, t = \partial f_3$. Compatibility therefore requires $\partial f_1 = 0$. If $\partial f_1 \neq 0$, there is no stationary horizontal solution. If $\partial f_1 = 0$, then $t = \partial f_3/(2 f_2)$ while q and u remain invariant algebraic moduli. The extra kernel directions are these horizontal moduli together with the vertical gauge V; they are not additional projective gauge.

S3, $f_1 \neq 0$, $f_2 = 0$. $\operatorname{rank}_{\mathrm{sp}}(P) = 2$, $\operatorname{rank}_{\mathrm{sp}}(H_\perp) = 2$, $\operatorname{null}_{\mathrm{sp}}(H_{\mathrm{full}}) = 2$. At $D = 4$ the fibre values are rank 8 and nullity 8. The Palatini block still determines (q, u) as functions of t. Substituting into the residue row yields the identity $-\partial f_3 = 0$ independently of t. Compatibility therefore requires $\partial f_3 = 0$. If $\partial f_3 \neq 0$, there is no stationary solution. If $\partial f_3 = 0$, then t is

a gauge-invariant modulus and (q, u) are determined by t. The extra kernel is this invariant t-direction, not additional gauge.

S4, $f_1 = f_2 = 0$. $H_\perp = 0$, species kernel dimension four, fibre kernel dimension 16 at $D = 4$. Compatibility is $J = 0$, i.e. $\partial f_1 = 0$ and $\partial f_3 = 0$. If either fails, there is no stationary solution. If both hold, (q, u, t) are horizontal moduli and V remains vertical gauge.

Alternative first-index shift under the present targets. The pair $(a, b) = (1, 0)$ induces the trace map $(3, 2, 5)\xi$. It fails the Ricci target (4.6) and is not selected by Proposition 4.1.

IBP remainder versus charges. The bulk density remains projectively invariant. Off shell, $\delta J_1 = \delta J_3 = 0$ for inert spectators. The condition $n_\mu \delta(J_1^\mu + J_3^\mu) = 0$ therefore cannot classify proper versus improper transformations. Test C of Appendix H lies outside the theorem.

Deleted irreps as unknowns. Axial torsion, tensor torsion, and traceless nonmetricity are not coordinates of H_{full}. Treating them as solved variables changes the domain rather than enlarging the kernel computation.

No additional algebraic factor of $\det H_\perp$ appears. The regular proof is not extended through S2–S4.

J Minimal deformation control of the residue fingerprint

The following deformations test the residue fingerprint. The projective law and horizontal chart are held fixed, while neighboring actions are defined by additions to the undeformed action (3.7). A local deformation is called *symmetry-compatible* when it is a diffeomorphism scalar constructed from the projectively invariant horizontal one-forms, at the same algebraic and derivative order as the undeformed quadratic density, and independent of the vertical coordinate V.

Write $x = (q, u, t)$ and isolate the Palatini block P, mixing B, and residue coefficient C in the $\frac{1}{2}x^{\mathsf{T}}Hx$ normalization,

$$B = f_1 \begin{pmatrix} -2/3 \\ 2/3 \end{pmatrix}, \qquad C = f_1\big(-\tfrac{4}{3}\big) + 2f_2, \tag{J.1}$$

so that the t-Schur coefficient is $S_t = C - B^T P^{-1} B$. The undeformed value is $S_t(0, 0) = 2f_2$, equivalently $\Delta_2 = 0$.

Consider the nearby symmetry-compatible quadratic deformation

$$\delta L_{\text{quad}} = \alpha\, q \cdot t + \beta\, u \cdot t, \tag{J.2}$$

with independent symbolic coefficients α, β. This shifts $B \mapsto B + \delta B$ with $\delta B = (\alpha, \beta)^{\mathsf{T}}$. Then

$$S_t(\alpha, \beta) = C - (B + \delta B)^T P^{-1} (B + \delta B). \tag{J.3}$$

Using (5.17) one finds $B^T P^{-1} \delta B = \frac{8}{3}\alpha + \frac{2}{3}\beta$ and

$$\delta B^T P^{-1} \delta B = f_1^{-1}\Big(-\tfrac{4}{3}\alpha^2 + \tfrac{16}{3}\alpha\beta + \tfrac{11}{3}\beta^2\Big). \tag{J.4}$$

Hence

$$\delta S_t := S_t(\alpha, \beta) - S_t(0, 0) = -\tfrac{16}{3}\alpha - \tfrac{4}{3}\beta - f_1^{-1}\Big(-\tfrac{4}{3}\alpha^2 + \tfrac{16}{3}\alpha\beta + \tfrac{11}{3}\beta^2\Big). \tag{J.5}$$

Because the Palatini block is unchanged and $f_1 \neq 0$, the block-determinant identity also gives

$$\det H_\perp(\alpha,\beta) = \det P\, S_t(\alpha,\beta) = -\frac{f_1^2}{12}\, S_t(\alpha,\beta).$$

For each $f_1 \neq 0$, the zero locus of (J.5) is a plane conic of measure zero in the (α,β) plane. Off that conic, $\delta S_t \neq 0$, so Δ_2 is corrected. In particular, the line $\beta = -4\alpha$ kills the linear part and yields $\delta S_t = -36\alpha^2/f_1$, which vanishes only at $\alpha = 0$.

The undeformed t-source Schur correction is $J_{t,\mathrm{eff}} = -\partial f_3$, equivalently $\Delta_3 = 0$. With (J.2) one obtains the extra term

$$J_{t,\mathrm{eff}}(\alpha,\beta) = -\partial f_3 + \frac{\partial f_1}{f_1}(4\alpha+\beta). \tag{J.6}$$

As a structural identity for arbitrary ∂f_1, preservation of the effective source requires $4\alpha+\beta = 0$. On a fixed special background with $\partial f_1 = 0$, however, the source correction vanishes independently of (α,β). For structural preservation of the Schur coefficient, multiplication of (J.5) by $3f_1$ gives

$$-(16\alpha+4\beta)f_1 + 4\alpha^2 - 16\alpha\beta - 11\beta^2.$$

Identity over the admitted coefficient-function class first requires $\beta = -4\alpha$; the remaining term is then $-108\alpha^2$. Hence simultaneous structural preservation of both corrections occurs exactly at $(\alpha,\beta) = (0,0)$, proving Proposition 9.1. Fixed-background zeros of (J.5) can contain additional accidental points; for example, $f_1 = c \neq 0$, $\partial f_1 = 0$, $\alpha = 0$, and $\beta = -4c/11$ preserve both corrections.

For the deformed-family result of Proposition 8.1, the residual equation is

$$S_t(\alpha,\beta)t - \partial f_3 + \frac{\partial f_1}{f_1}(4\alpha+\beta) = 0.$$

Thus $S_t \neq 0$ gives a unique residue solution. On $S_t = 0$, solvability requires the displayed source compatibility condition; when it holds, t remains free and the invertible Palatini block determines (q,u) affinely from it. This is one invariant horizontal modulus in addition to the vertical projective direction.

A minimal symmetry-compatible linear-source deformation $J_N \mapsto J_N + (\varepsilon_q, \varepsilon_u)$ produces, at $\delta B = 0$,

$$\delta J_{t,\mathrm{eff}} = -\left(\tfrac{8}{3}\varepsilon_q + \tfrac{2}{3}\varepsilon_u\right). \tag{J.7}$$

This vanishes if and only if $\varepsilon_u = -4\varepsilon_q$. A generic source deformation therefore yields a nonzero t-source Schur correction.

The fingerprint $\Delta_2 = \Delta_3 = 0$ is structurally non-generic within the stated deformation family. The undeformed model remains (3.7); Proposition 8.1 separately treats the neighboring actions defined by (J.2).

Computational reproducibility. The identities for $S_t(\alpha,\beta)$, $J_{t,\mathrm{eff}}(\alpha,\beta)$, the determinant factorization, the rank-two Schur surface, its null vector, the undeformed limit, and the fixed-f_1 counterexample were checked with exact rational arithmetic in Python 3.7.3 using the standard-library `fractions.Fraction` type. The standalone supplementary file `JGP_algebra_verification.py`, submitted with this manuscript together with its README, also performs the independent component curvature and trace-inversion controls of Appendix C. Cursor AI assisted preparation of the verification code as disclosed in the manuscript-preparation statement; the displayed derivations and identities were independently reviewed by the author, and the code output is not used as a substitute for proof.

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Data availability

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Cursor AI to support language editing, submission engineering, literature discovery, consistency checks, and preparation of exact-arithmetic verification code for Appendices C and J and the associated algebraic analyses. The author reviewed and edited the resulting material, independently reviewed the displayed derivations, remains responsible for all derivations, citations, and conclusions, and takes full responsibility for the content of the article.

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